



Unit 1

PROSPERITY, INEQUALITY AND PLANETARY LIMITS

Unit 1 Coverage

- Economic inequality and divergence
- The technological revolution and growth
- The role of capitalism in economic growth
- Importance of the government in capitalist economies
- The interaction between economics and the biosphere

Understanding economic inequalities

- We compare the countries, but on what grounds?
- How do we know that Singapore is *rich* and Uganda *poor*?

Measuring living standards

- *Gross Domestic Product* measures an economy's size:
 - the market value of all goods and services produced in a country within a given time period (typically one year).
 - measure of the total income earned within a country.
- *GDP per capita* measures average living standards:
 - income per person or $\frac{\text{GDP}}{\text{population}}$



GDP
GROSS
DOMESTIC
PRODUCT

GDP

GROSS DOMESTIC PRODUCT



Is GDP per capita a good measure of wellbeing?

Does GDP measure wellbeing?

“Our Gross National Product, now, is over \$800 billion dollars a year, but that Gross National Product ... counts air pollution and cigarette advertising, and ambulances to clear our highways of carnage. It counts special locks for our doors and the jails for the people who break them. It counts the destruction of the redwood and the loss of our natural wonder in chaotic sprawl... Yet the gross national product does not allow for the health of our children, the quality of their education or the joy of their play... It measures neither our wit nor our courage, neither our wisdom nor our learning, neither our compassion nor our devotion to our country, it measures everything in short, except that which makes life worthwhile.”



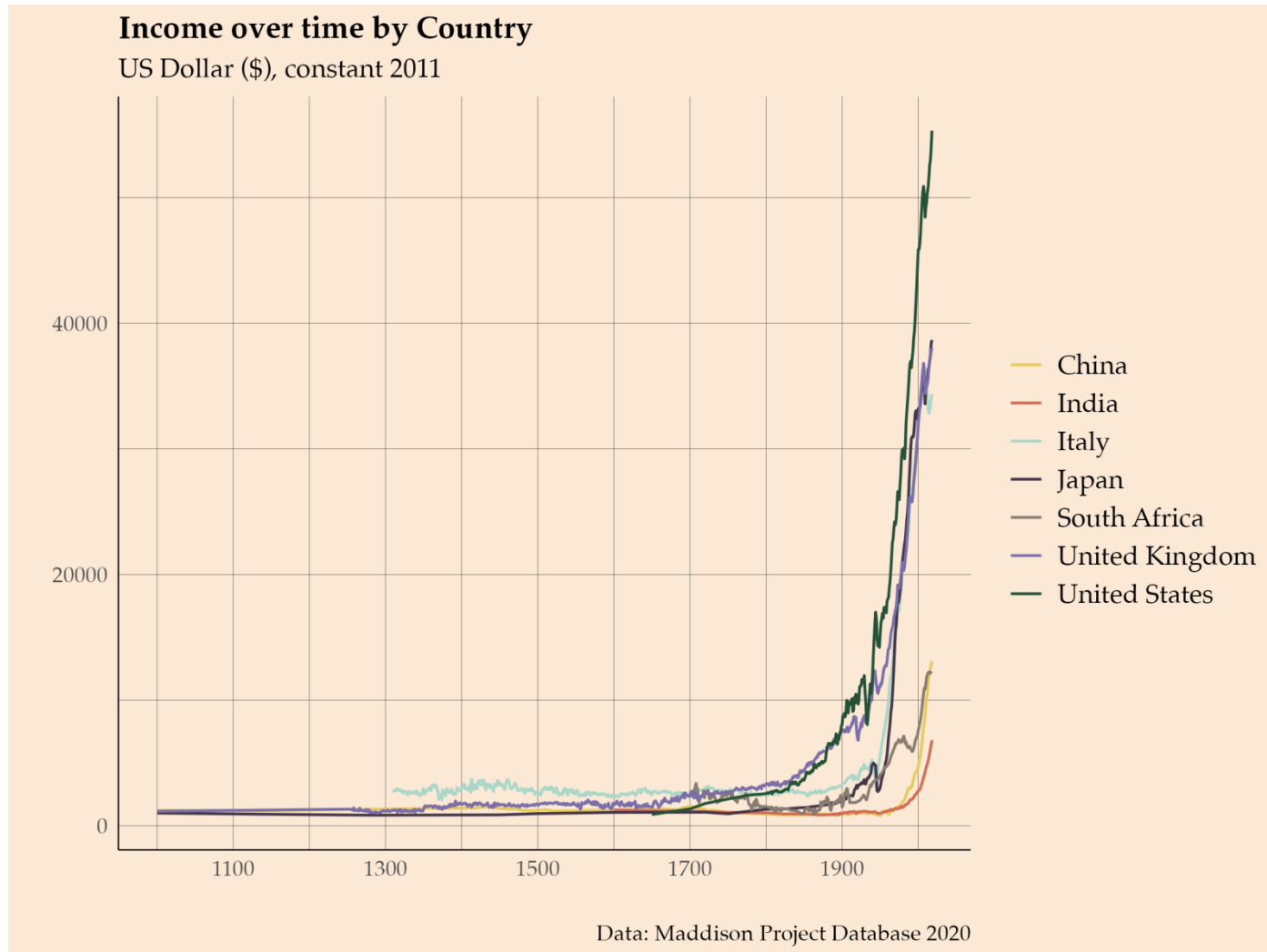
Robert F Kennedy, 1968

Can you think of other activities which GDP does not measure?

How unequal is the world?



Incomes across countries and over time



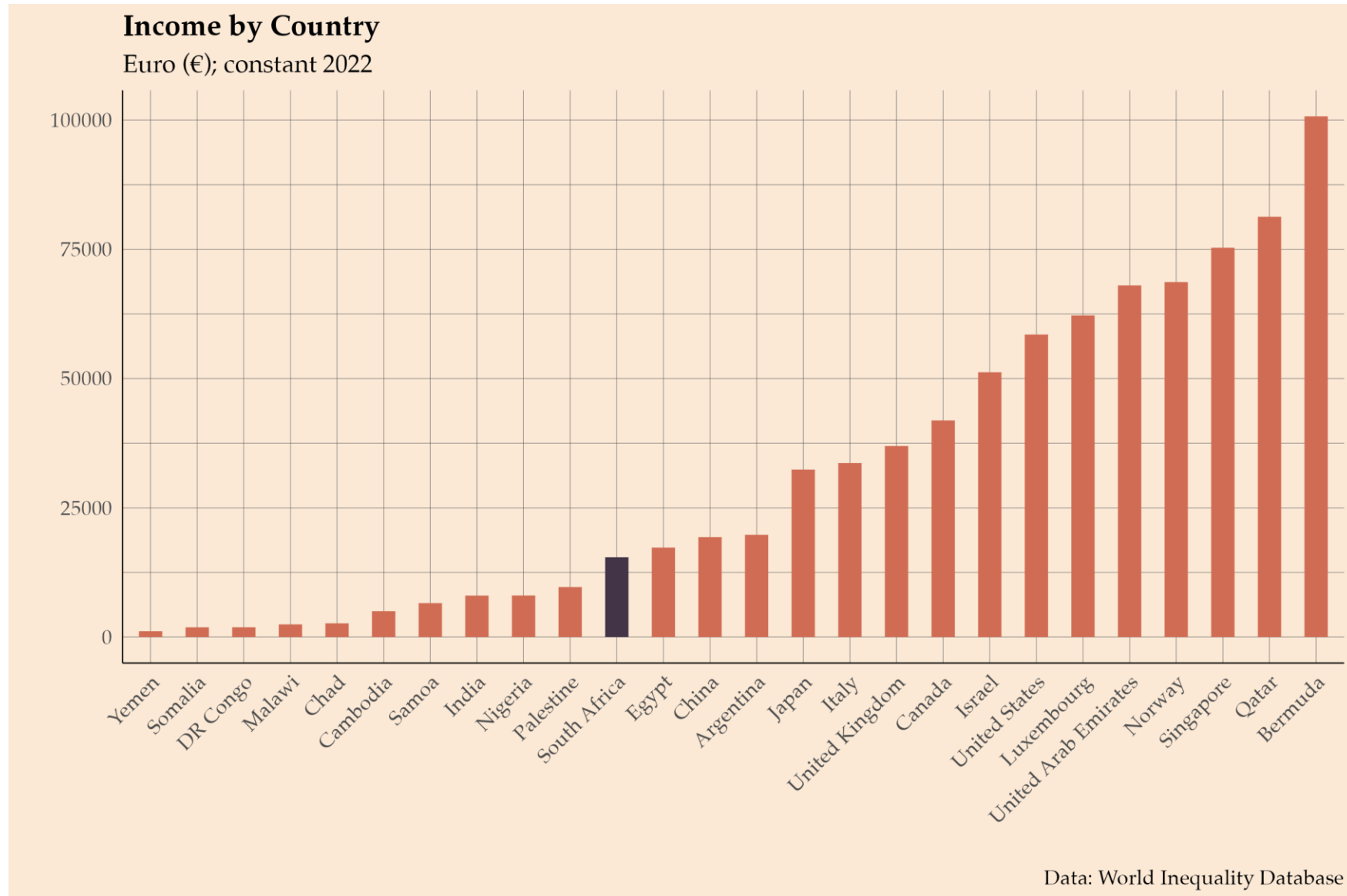
- A figure of income levels over time resembles a hockey stick.
- Relative incomes change:
 - In 1300, Italy was richest
 - In 1800, Britain was richest
 - From 1900, the US is even richer



Homework

- Think of any country
- Search on the [World Bank data bank](#):
 - *Country name* GDP
 - Write it down
 - *Country name* population
 - Write it down
 - Let's compare the countries
 - *Country name* GDP per capita

Between country inequality



Within country inequality

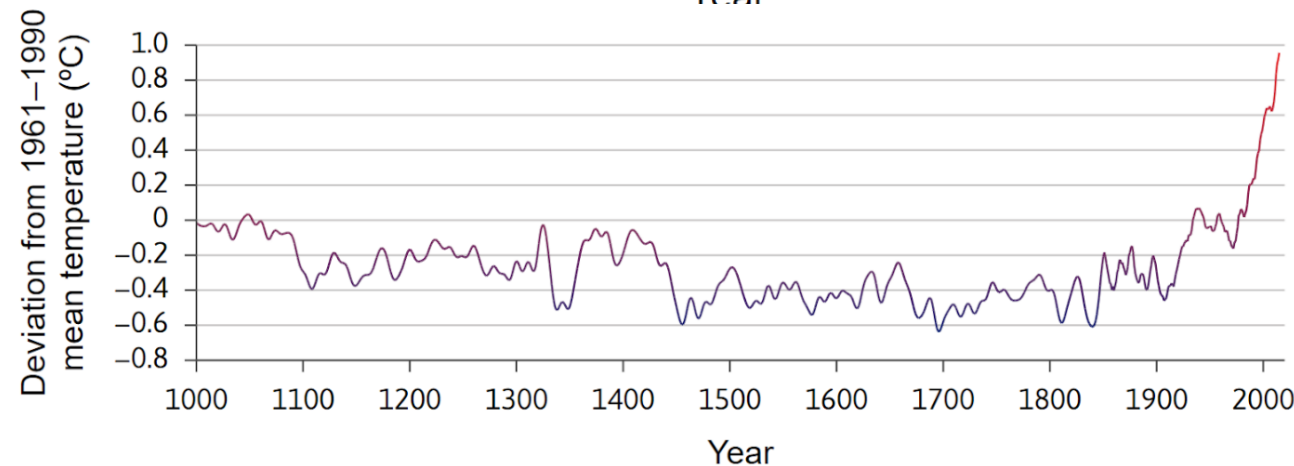
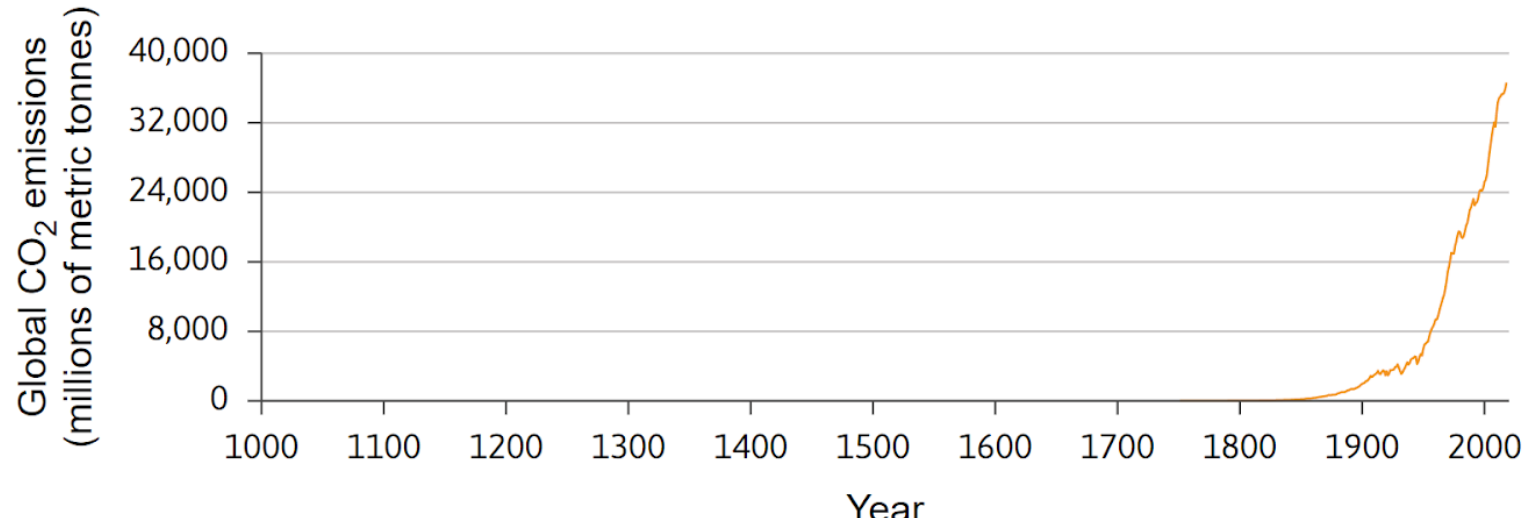


Share of Income in South Africa (2022)

1.2



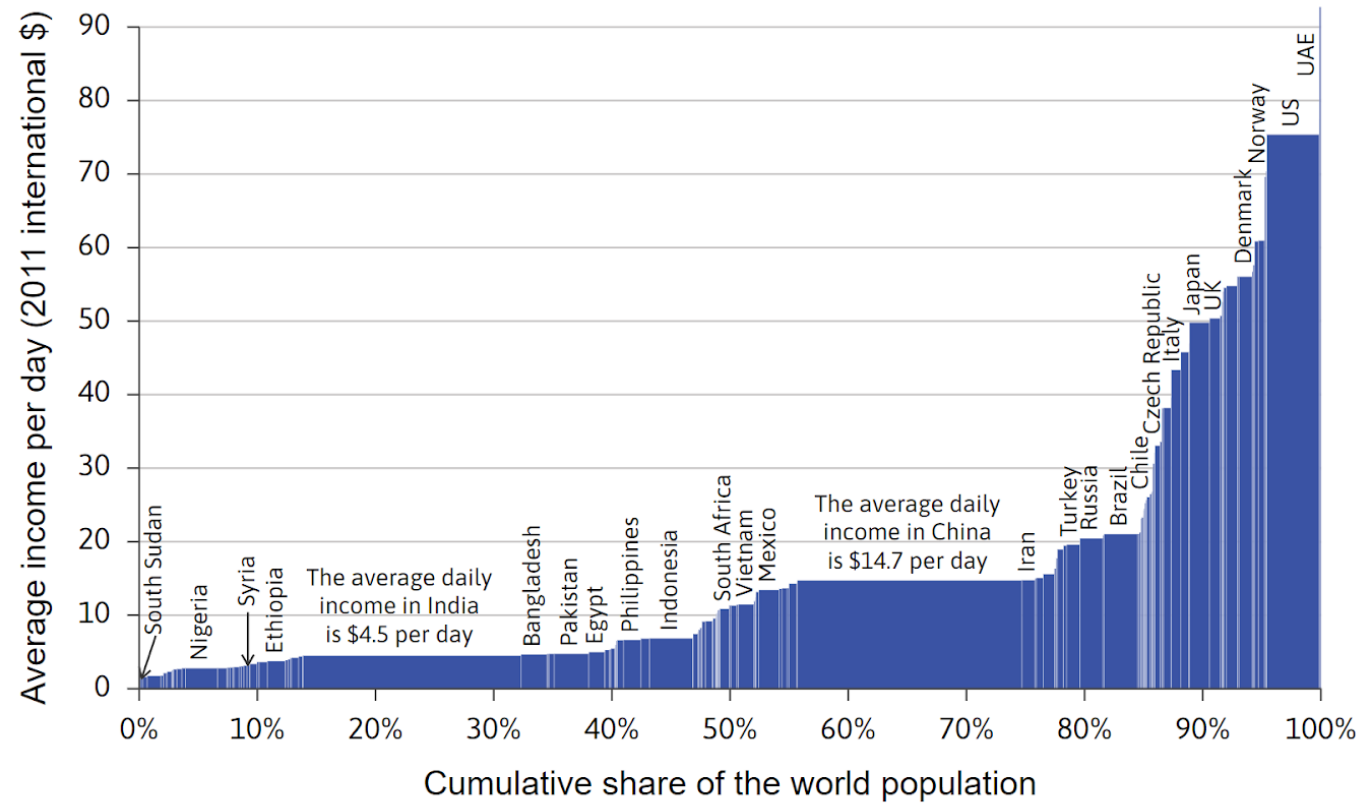
Cost of economic development



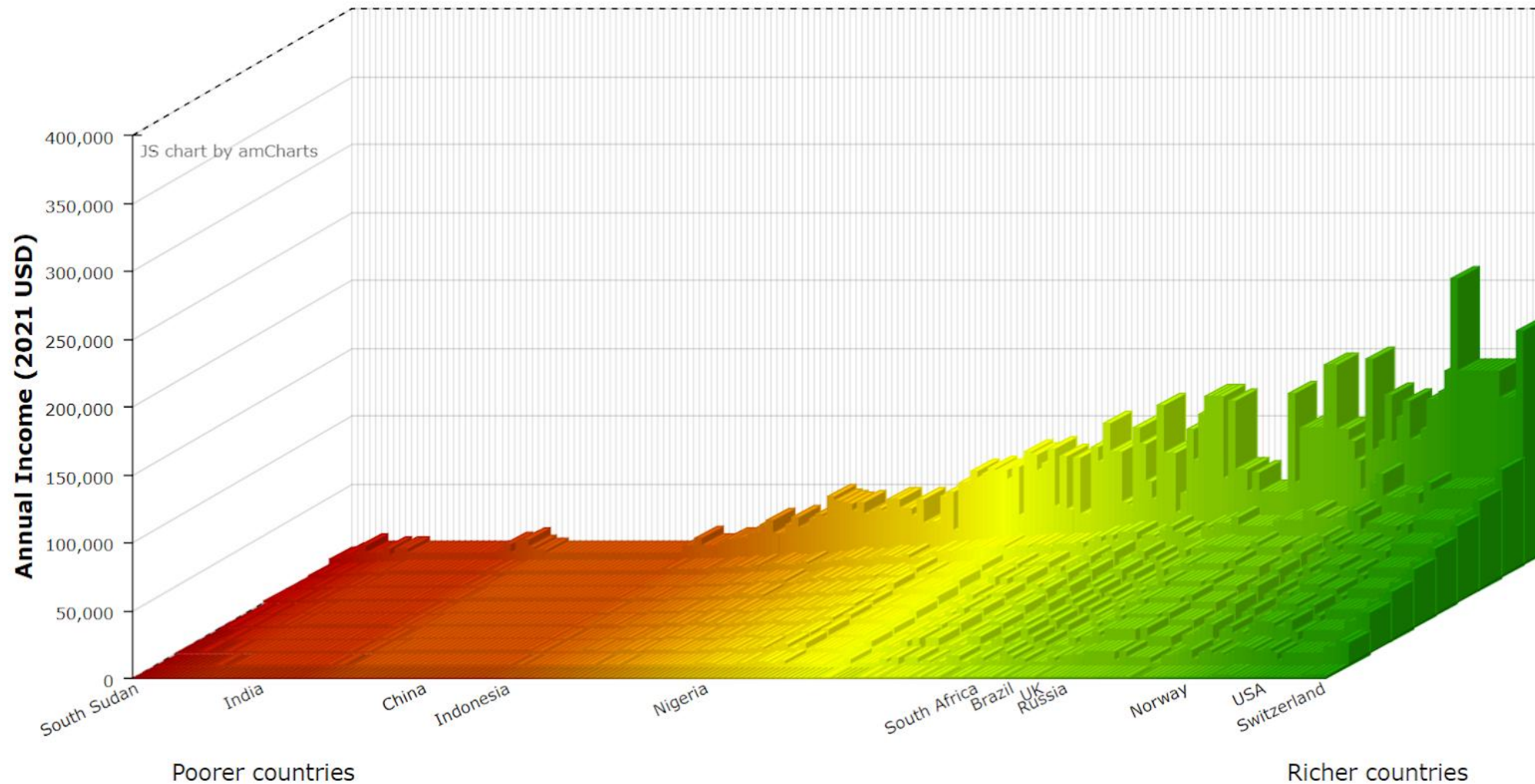
Consequences of climate change:

- Melting of polar ice caps
- Rising sea levels
- Changes in climate and rain patterns
- Destruction of food-growing areas
- Uninhabitable zones

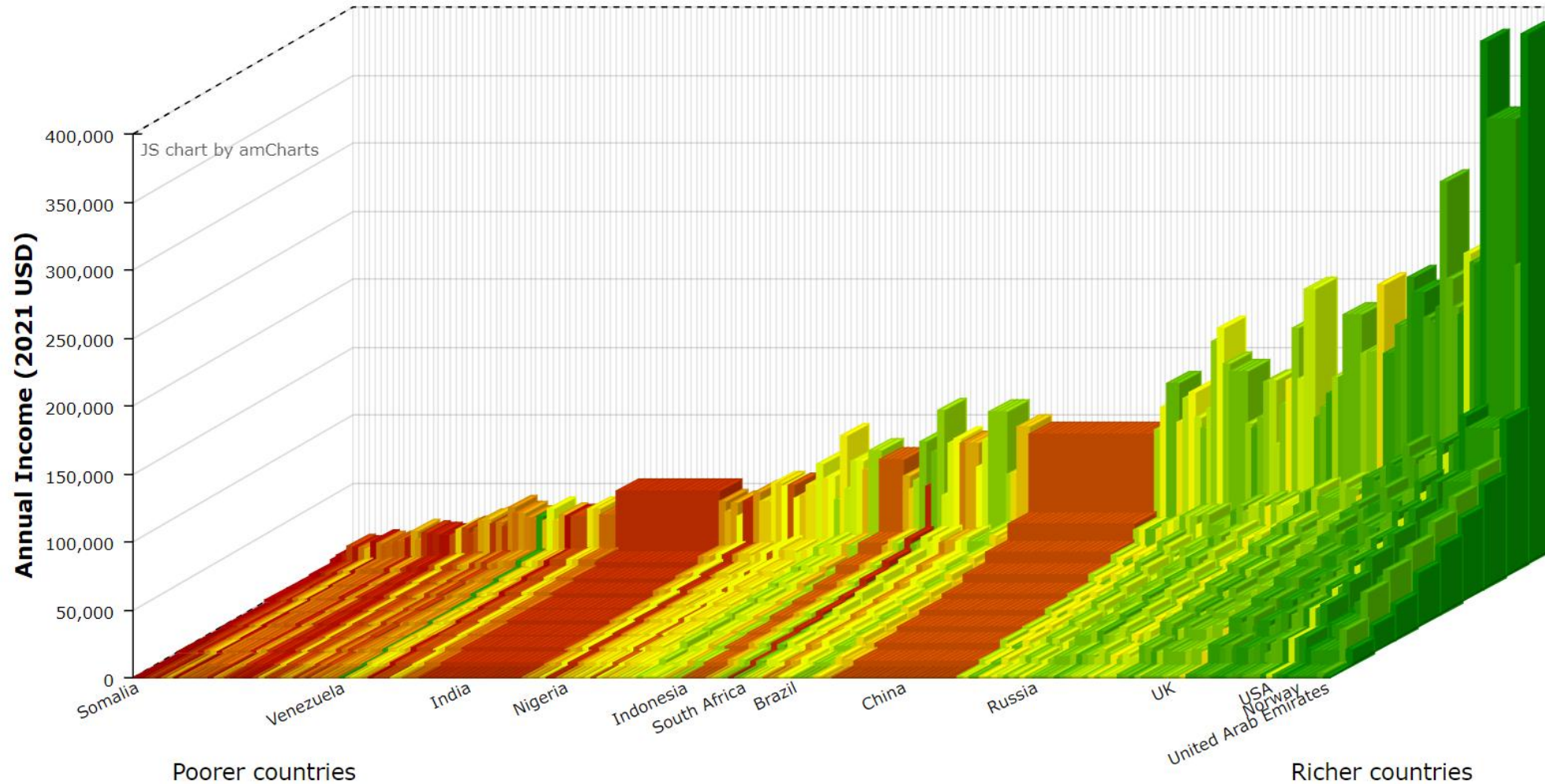
Income inequality

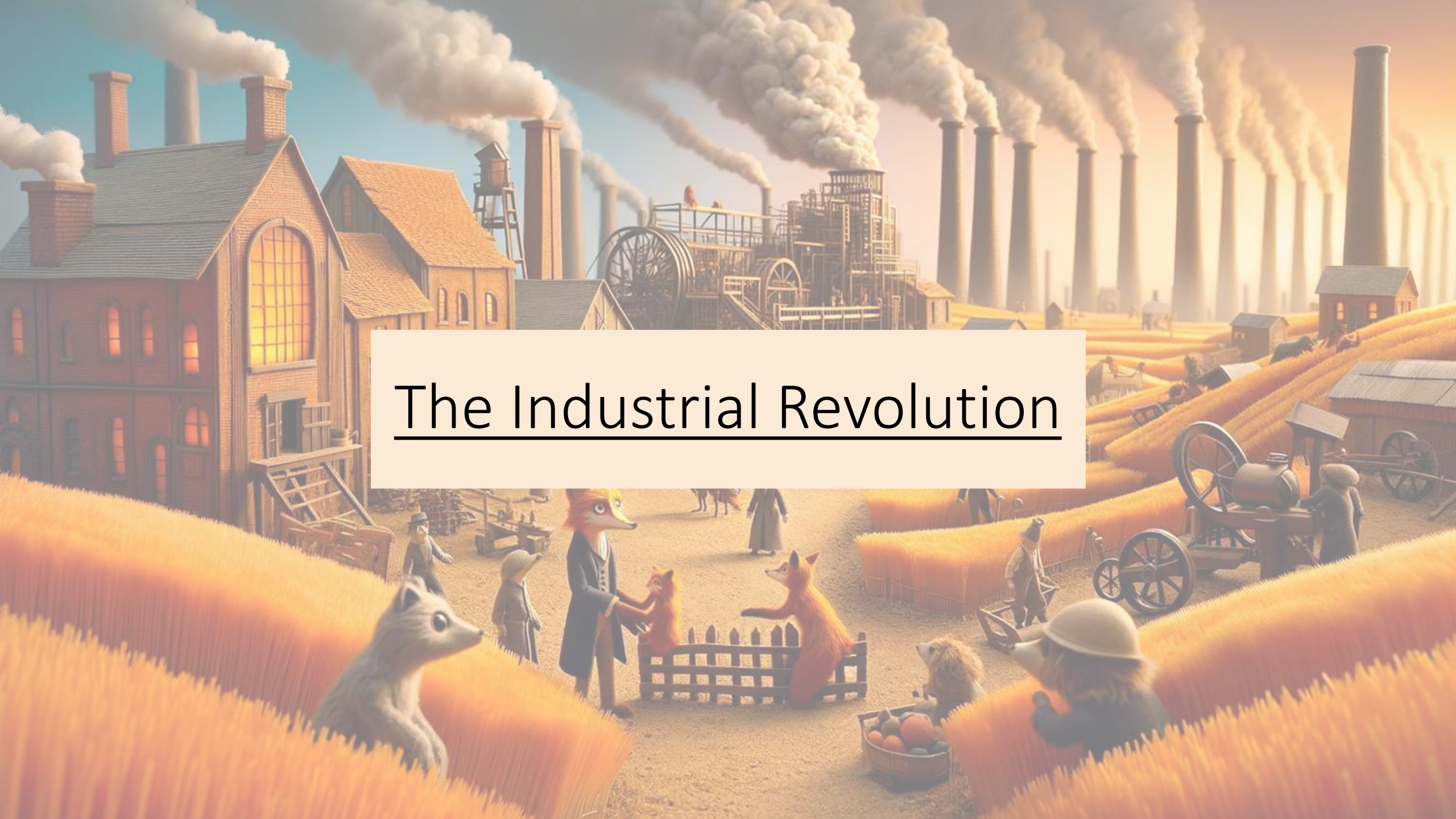


Income inequality (1980)



Income inequality (2020)



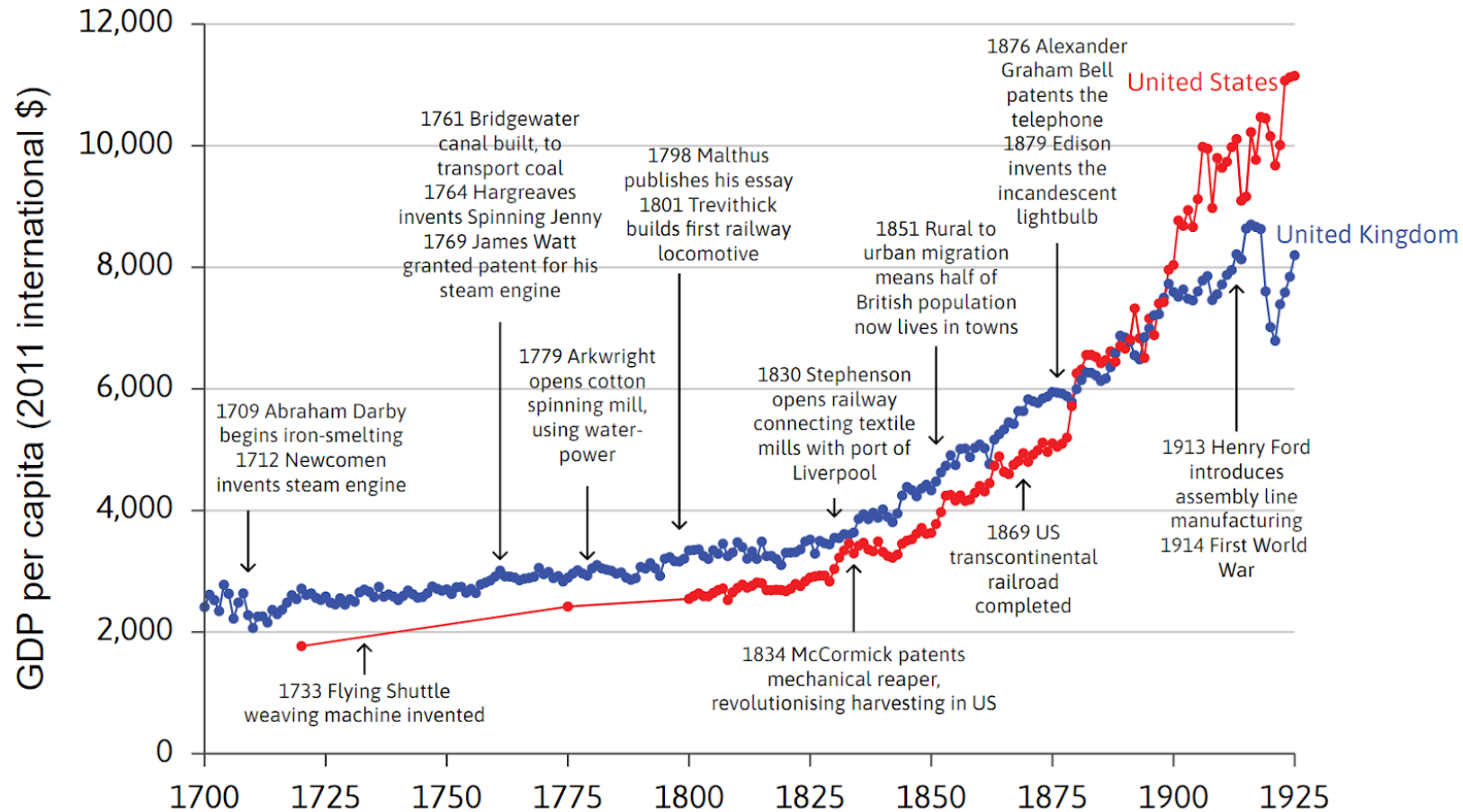


The Industrial Revolution

Technological Revolution

- Technology: Process of using inputs to create output
- Industrial Revolution
 - Technological progress
 - Decreased time to produce goods and services
 - Steam engine, spinning jetty, railroads, automobiles, etc.
 - Continuous process (with currently digital communication)

Fig 1.7 Examples of continuous technological revolution



Measuring economic growth

$$\text{Growth rate} = \frac{\text{change in GDP}}{\text{original level of GDP}}$$

$$= \frac{y_{2001} - y_{2000}}{y_{2000}}$$

$$= \frac{\$6,017 - \$5,938}{\$5,938}$$

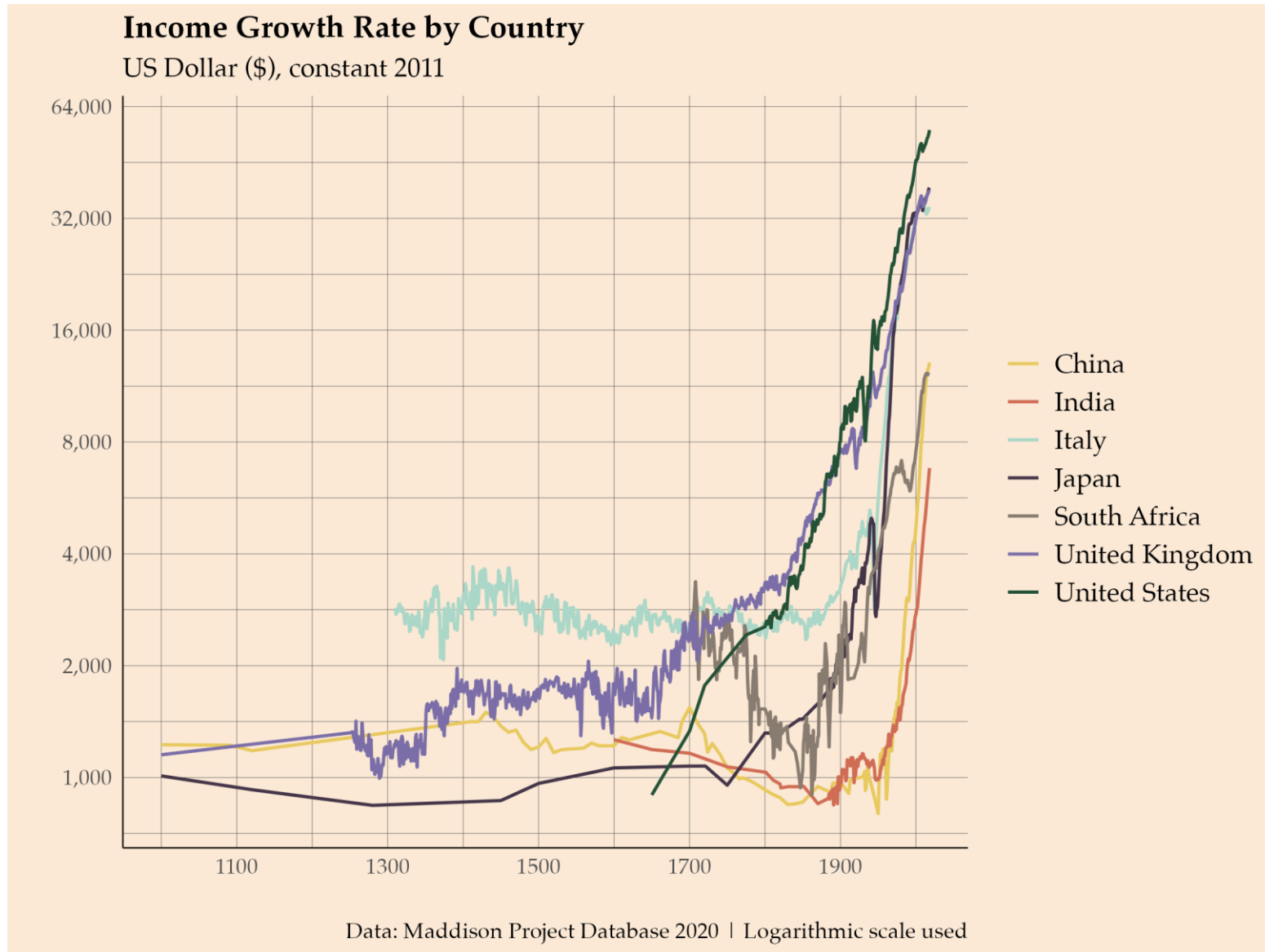
$$= 0.0133$$

$$= 1.33\%$$



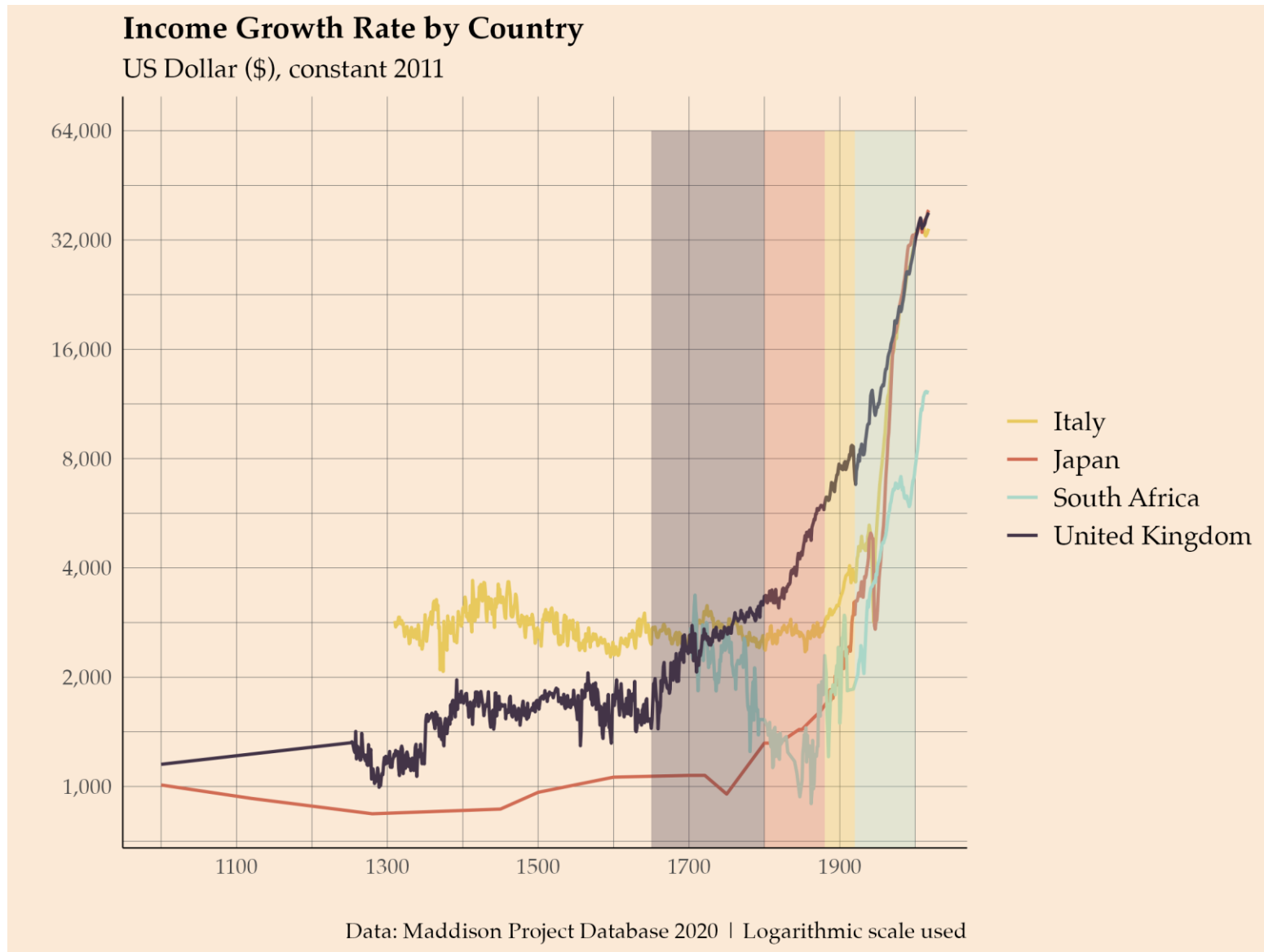
South African GDP per capita was \$5,938 in the year 2000 and \$6,017 in 2001.

GDP growth rates



“Hockey-stick” curves represent the sustained rapid growth in GDP per capita experienced by countries worldwide.

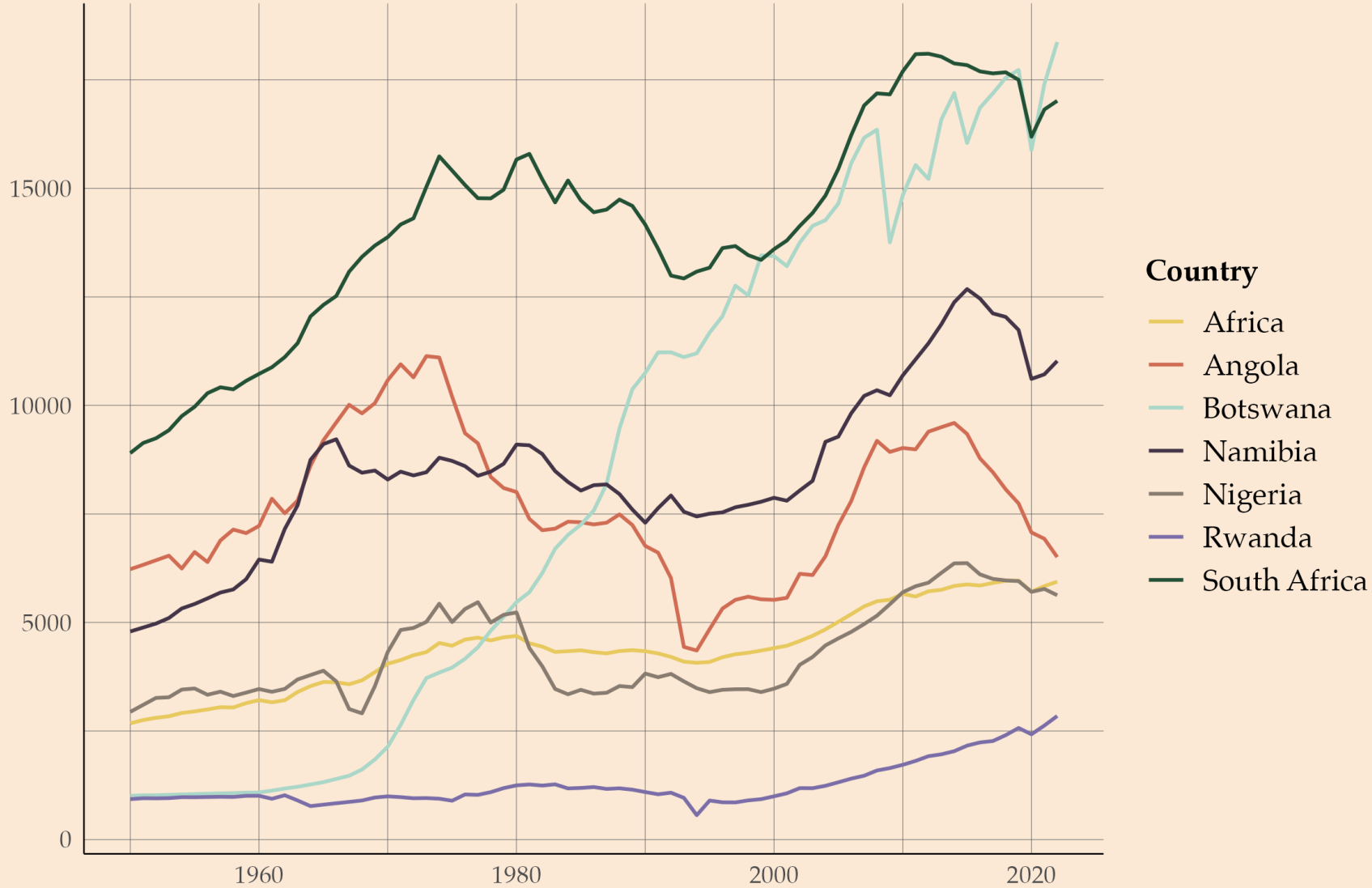
GDP growth rates



Timing of sustained growth occurred at different points in time

GDP Per Capita of African Countries

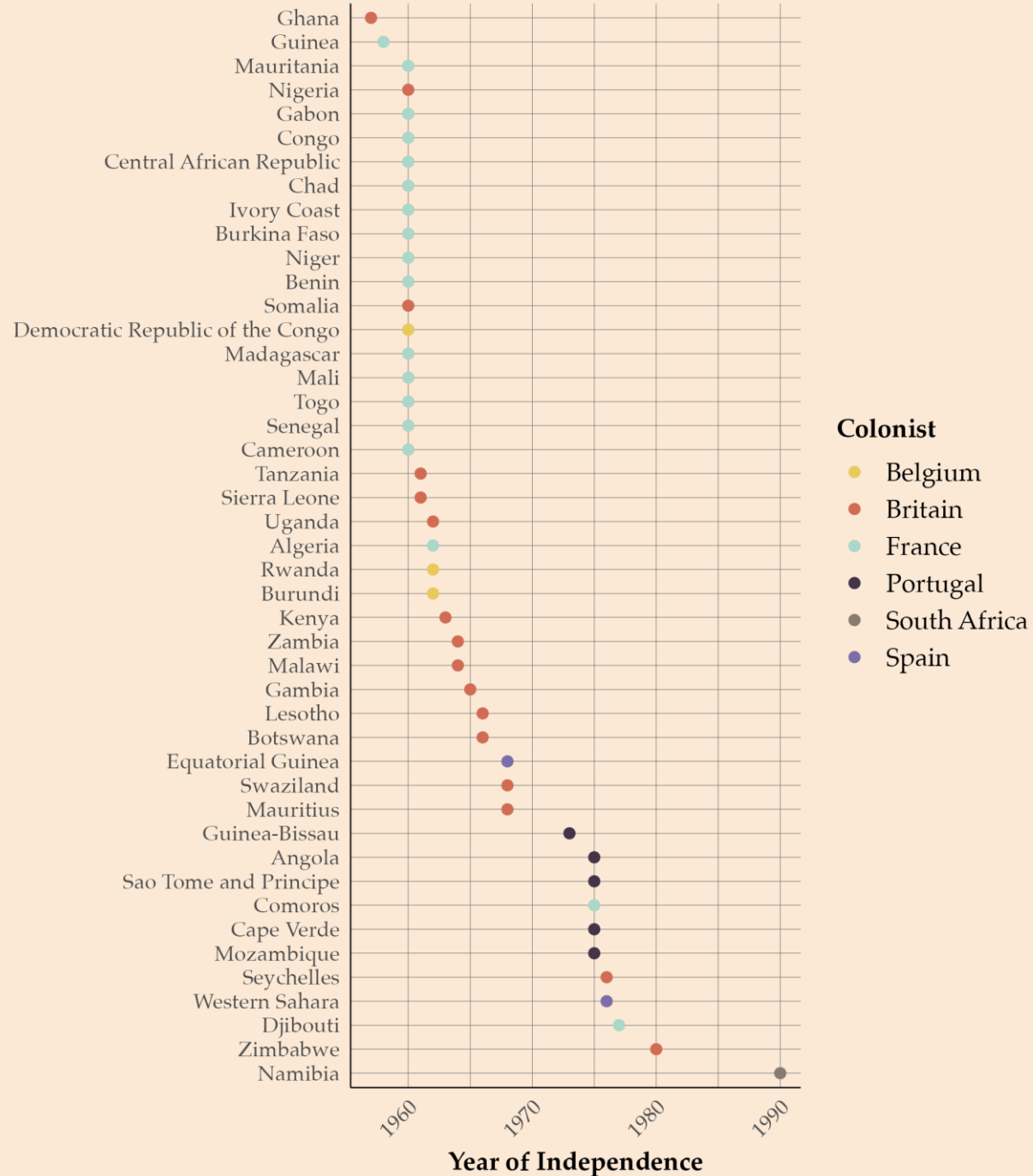
Constant US Dollar (\$) 2022



Data: World Inequality Database

Is there an
African
hockey
stick?

Timeline of African Countries' Independence



The African Hockey Stick?

- **Years of slavery and colonial rule**
- **Extractive Institutions**
- We might be seeing Africa's "takeoff", but this is not necessarily the same story as 1700 Europe.
- Mixed results ("takeoff" later in some countries e.g. Rwanda)
- Lack of quality data to measure economic growth
 - Why does the graph only start at 1950?

Timing of growth

Growth take-off occurred at different points in time for different countries:

- Britain was the first country to experience sustained economic growth. It began around 1650.
- In Japan, it occurred around 1870.
- The kink for China and India happened in the second half of the 20th century.

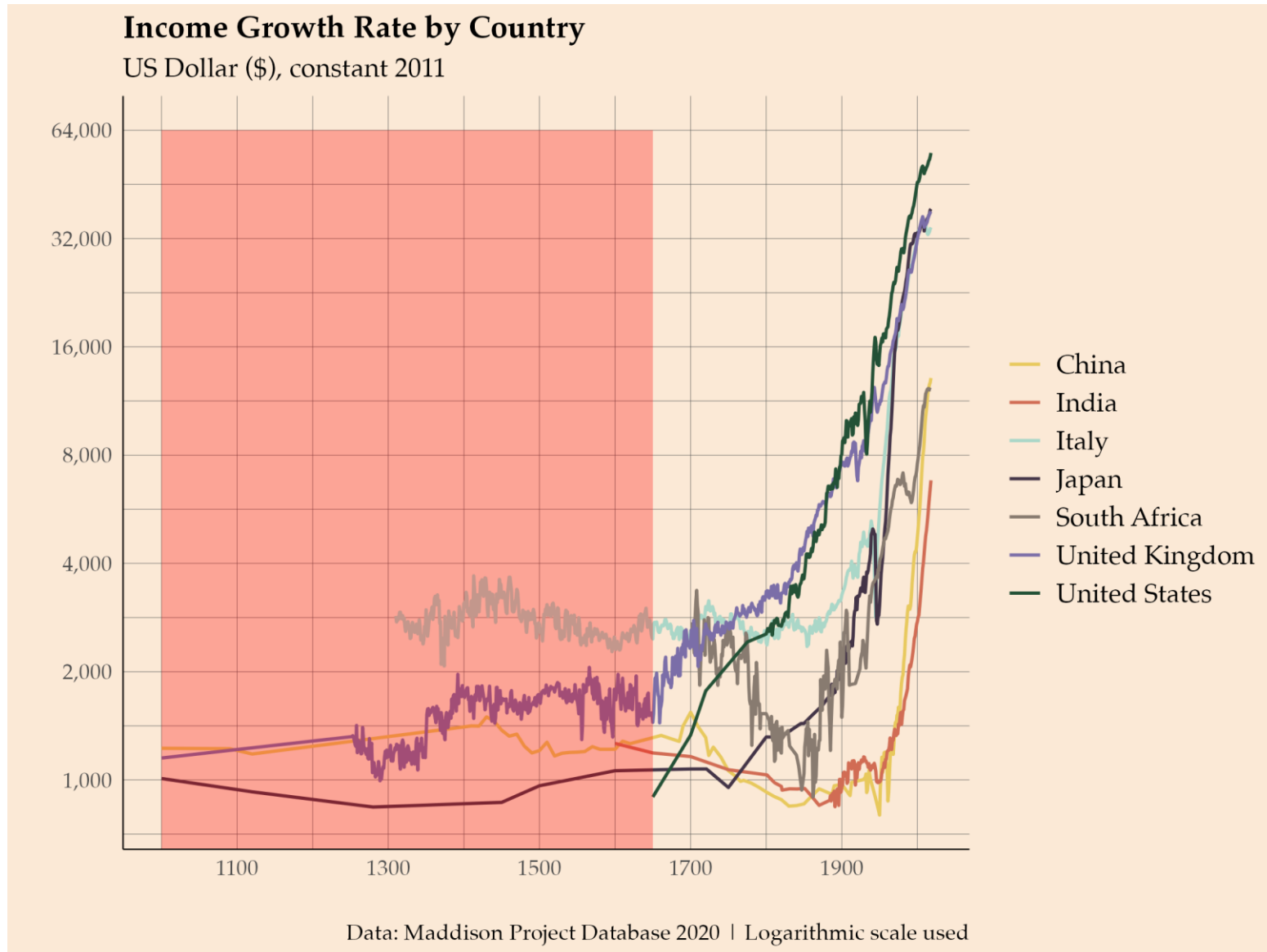
In some economies, substantial improvements in people's living standards did not occur until they gained independence from colonial rule or interference by European nations.

Inequality and growth

For a very long time, living standards did not grow in any sustained way. When sustained growth occurred it began at different times in different places.

- The countries that took off economically a century or more ago—UK, Japan, Italy—are now rich.
- The countries that took off only recently, or not at all, are in the flatlands. **Why do we have such a long period of low growth rates?**

GDP growth rates

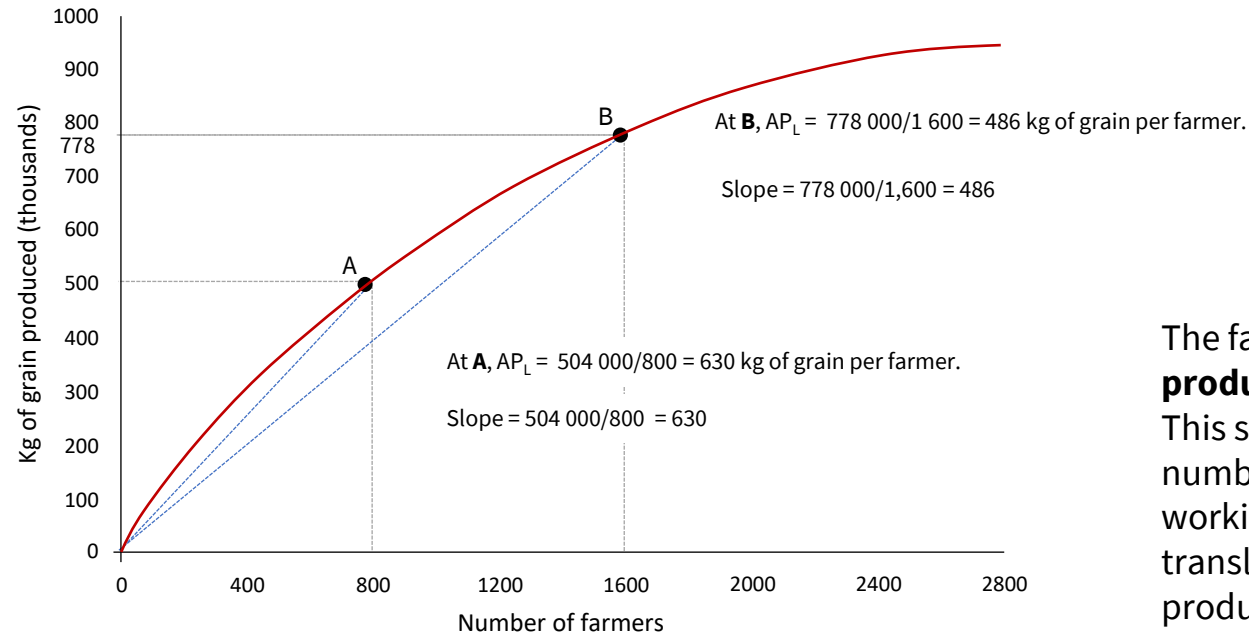


No sustained growth in living standards for close to seven centuries

Employment and Output

- Production function: Use inputs to convert them into output
- Agriculture
 - Assume land is fixed and only uses labour
 - Increasing labour means more workers on the same quantity of land
 - Average product of labour decreases ($AP_L \downarrow$)
 - $AP_L = \text{Total output} / \text{number of workers}$

Figure 1.8 – Production function and diminishing average product of labour



The farmer's **production function**:
This shows how the number of farmers working the land translates into grain produced.

Diminishing average product of labour

- As the number of farmers increases, the more grain is produced
- As more farmers work on a fixed quantity of land, the average product of labour decreases (See Fig 1.8)
- Earlier generations would have used the best land
- As production increases use more inferior land

Malthus: Population and living standards

- Population expands if living standards increase.
- Further population growth would push down incomes as a result of diminishing average product of labour.
- Falling living standards would slow population growth.
- Ultimately incomes would settle at a **subsistence level**. This is a level of income where the population remains constant over time.

Figure 1.10 – Equilibrium in a Malthusian model of an agricultural economy

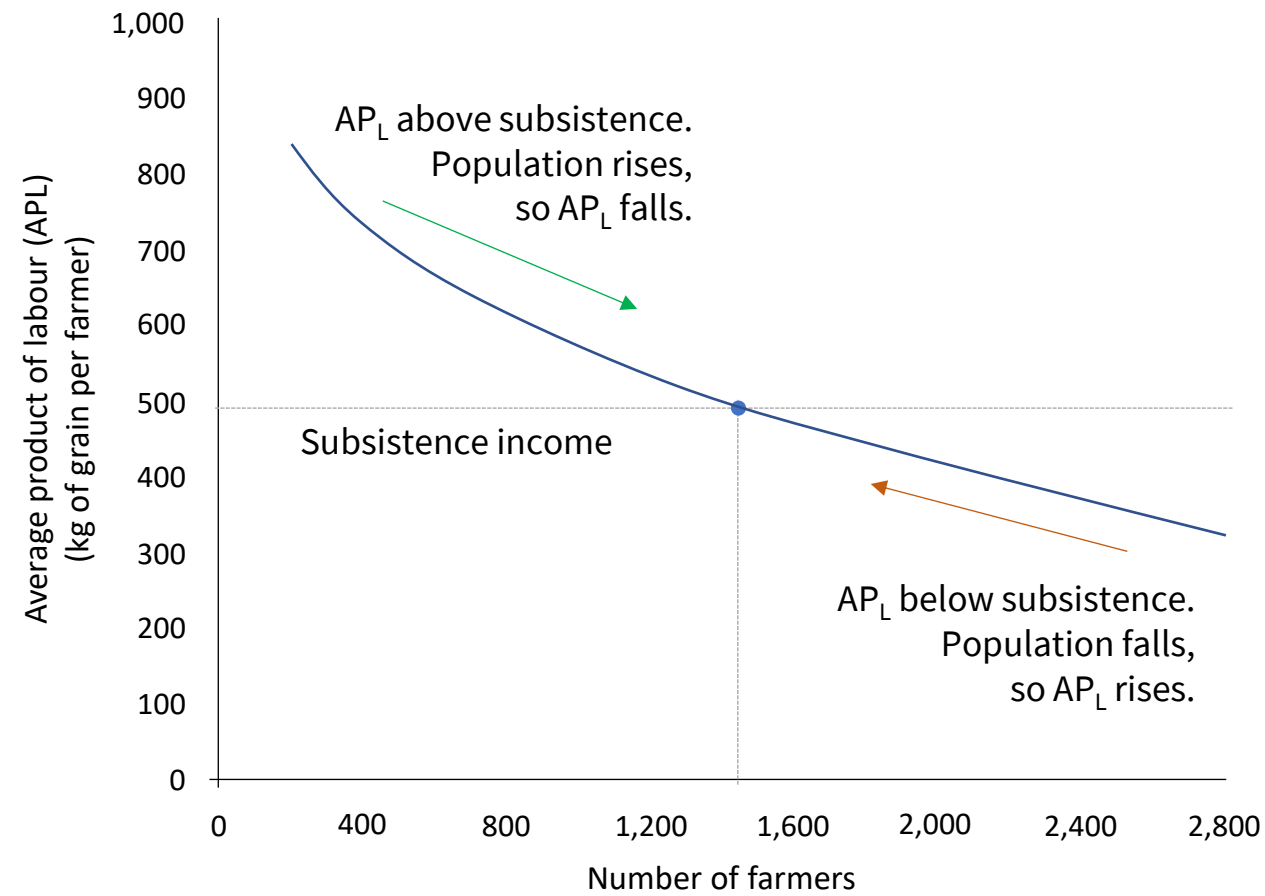


Figure 1.11 - Malthus model: The effect of an improvement in technology

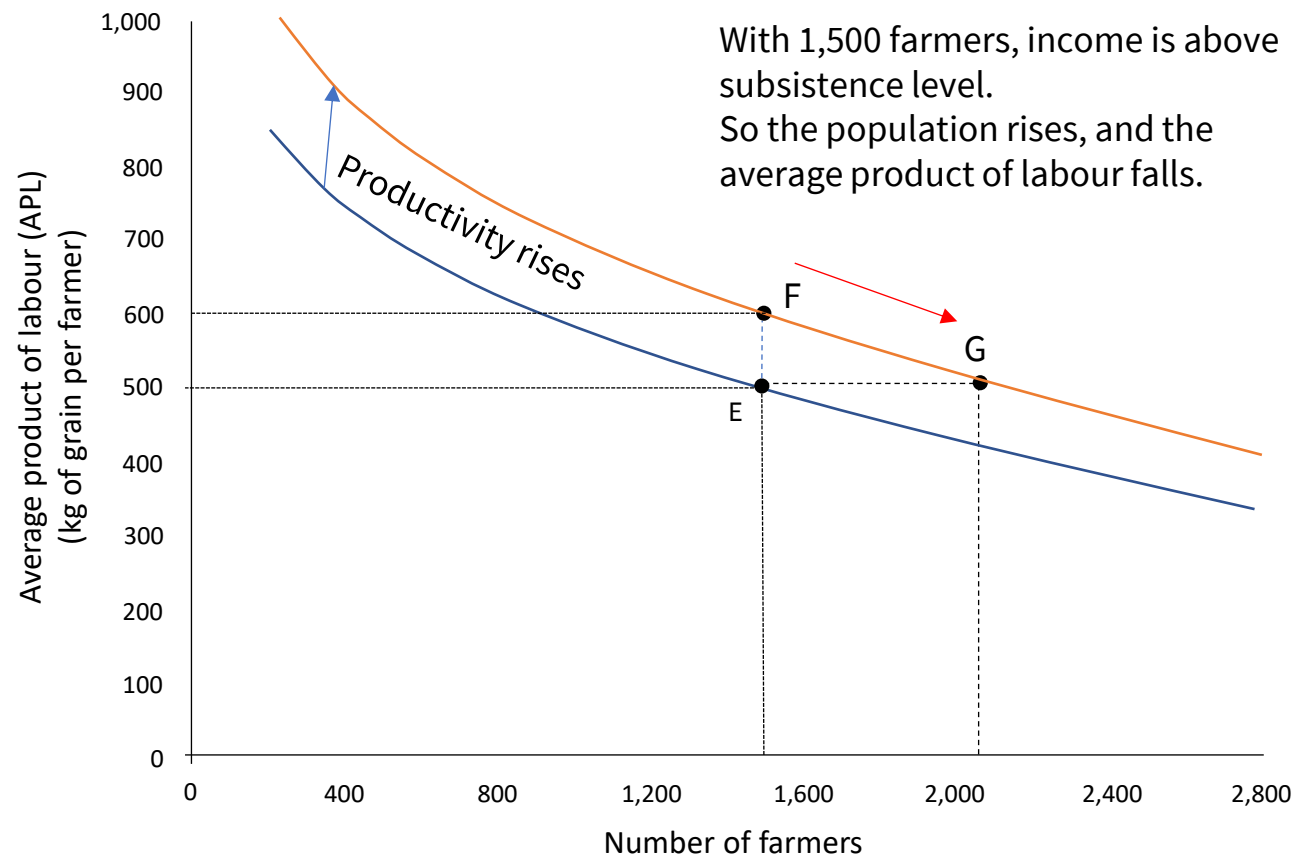
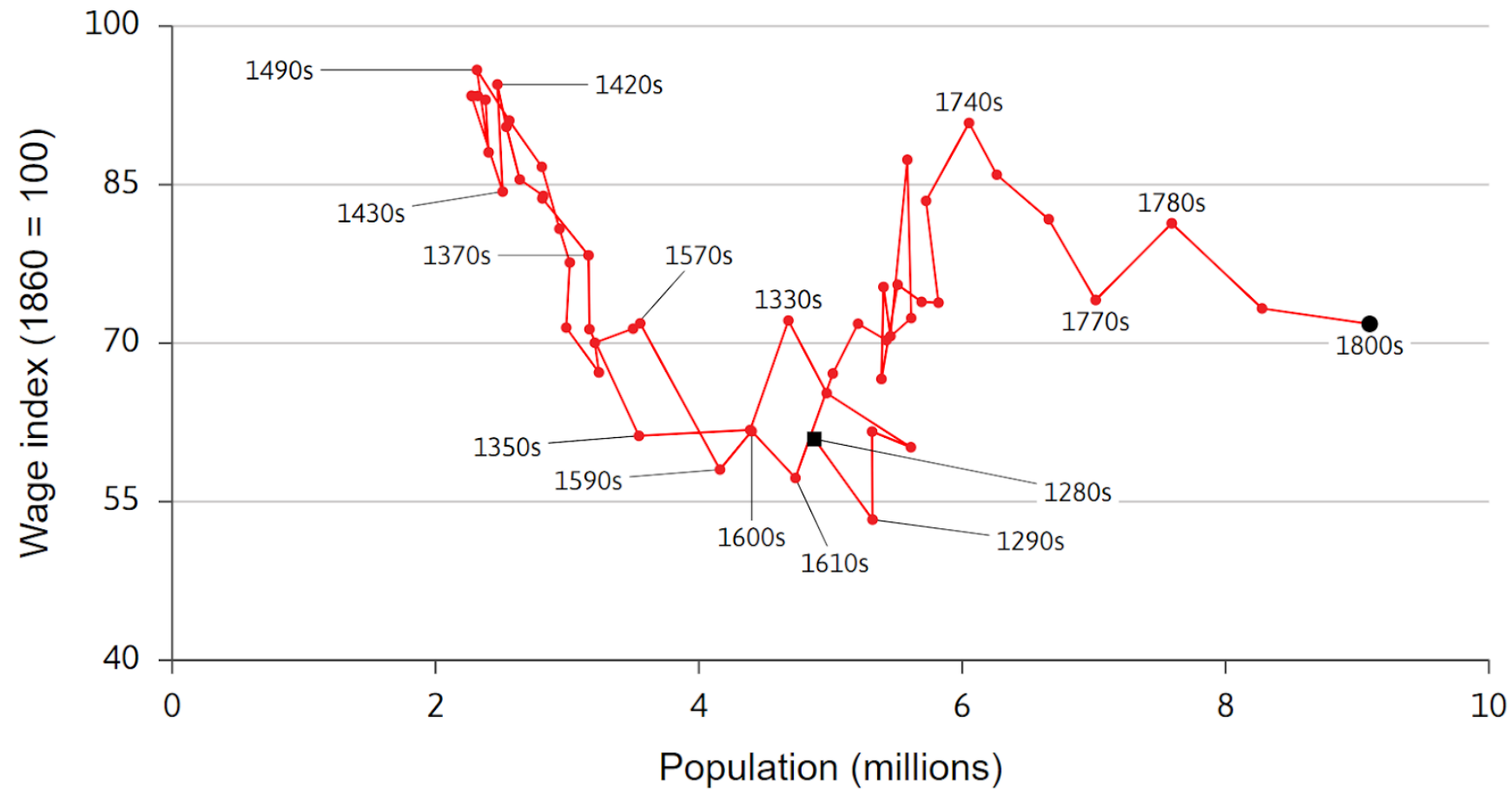
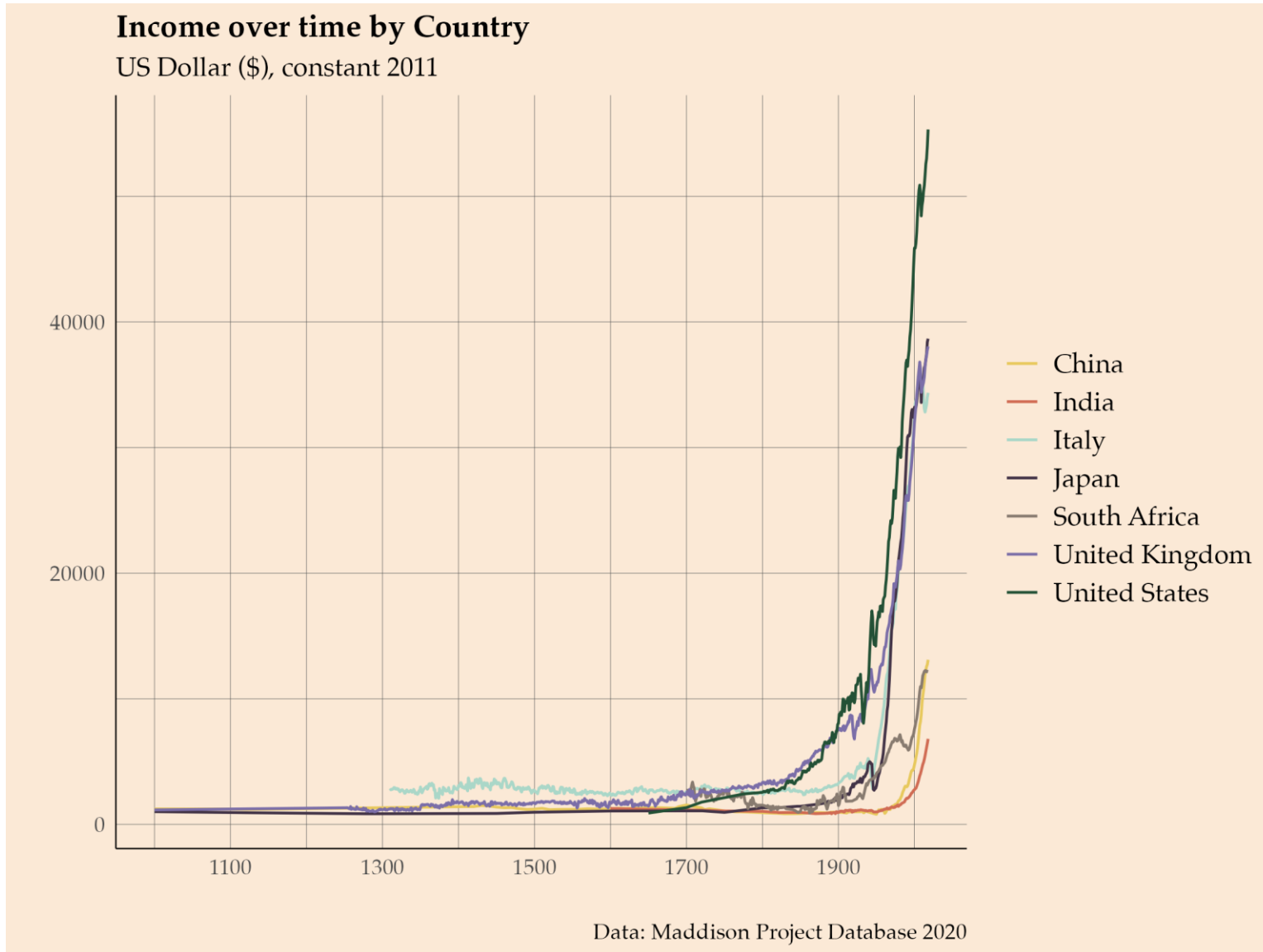


Figure 1.12: Evidence of Malthus' Law



The Context for This Lesson



Rapid, sustained growth in average living standards **since 1700.**
How did this happen?

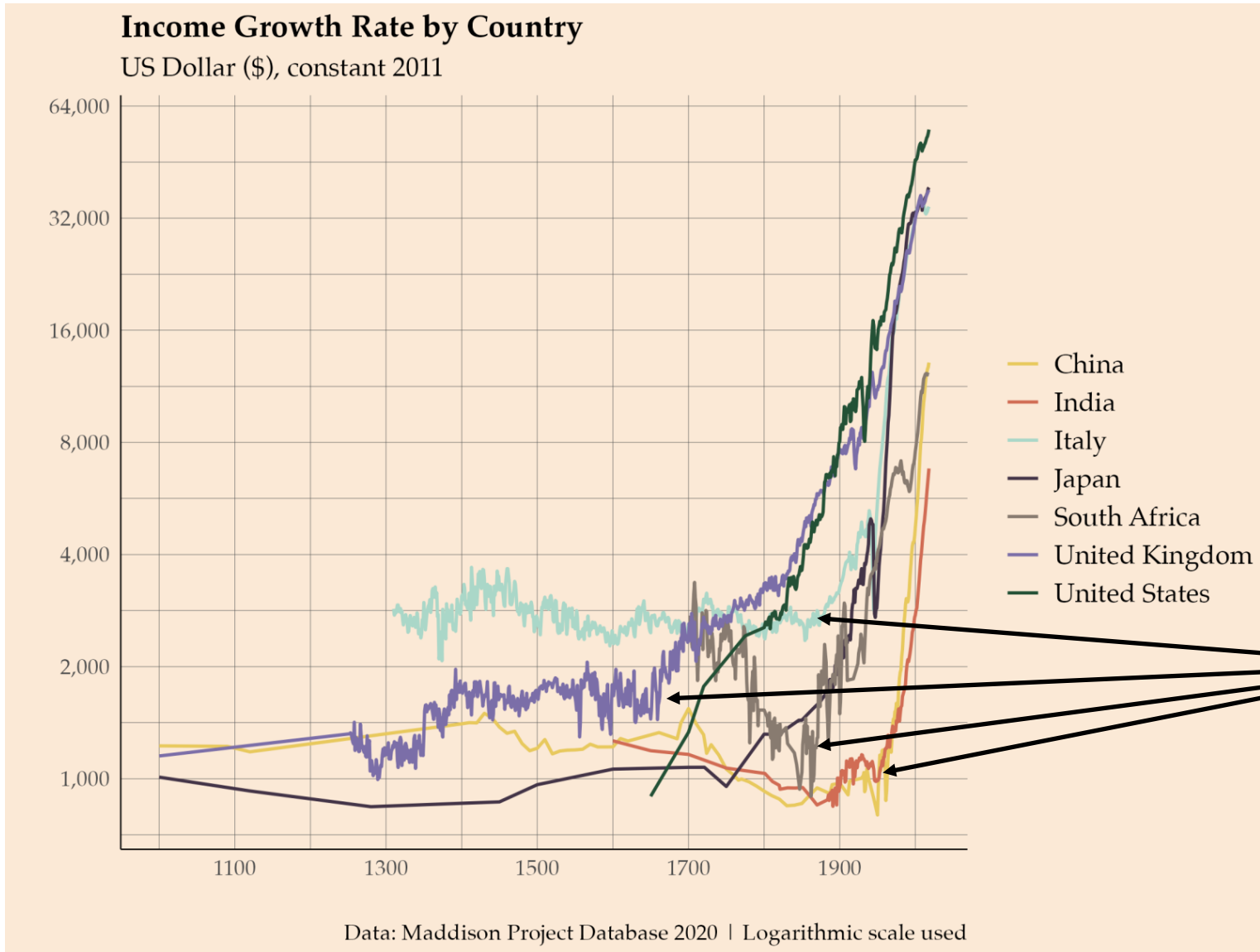
Capitalism: the permanent technological revolution

- A *technology* is a process that uses a set of inputs, including materials, machinery, and labour, to produce an output.
- Technological progress means less labour is needed to produce the same output.
- For millennia technological progress was very slow.
- Since the Industrial Revolution we have witnessed a *permanent technological revolution*



A washing machine represents tremendous technological progress

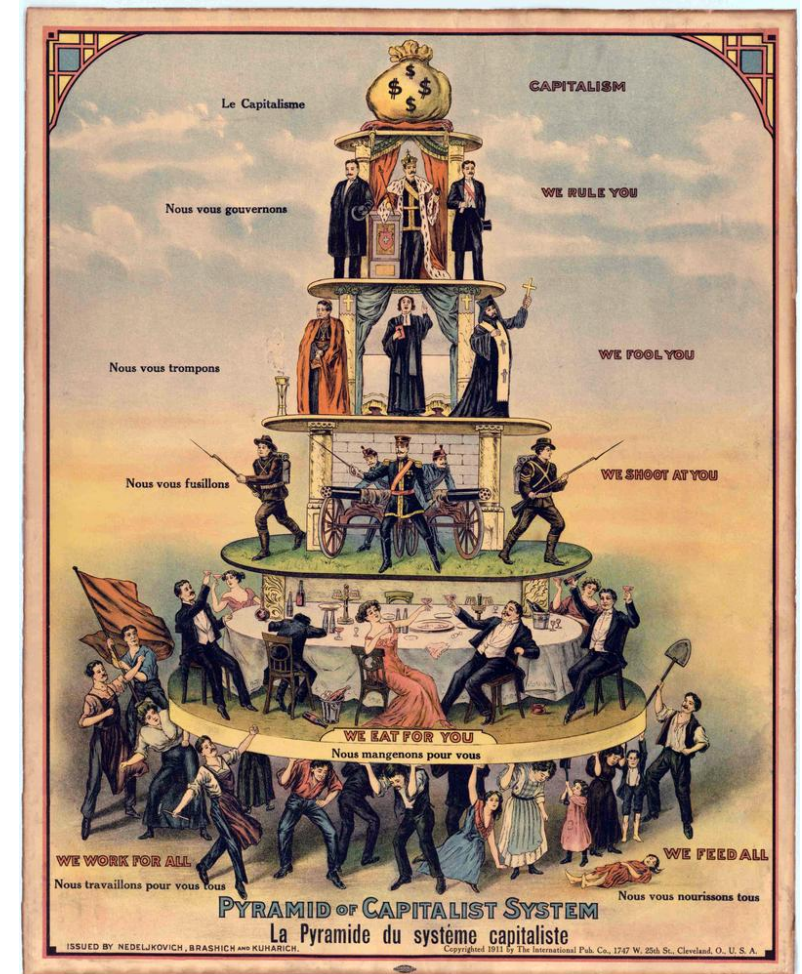
What explains the hockey sticks in GDP growth and income?



Capitalism?

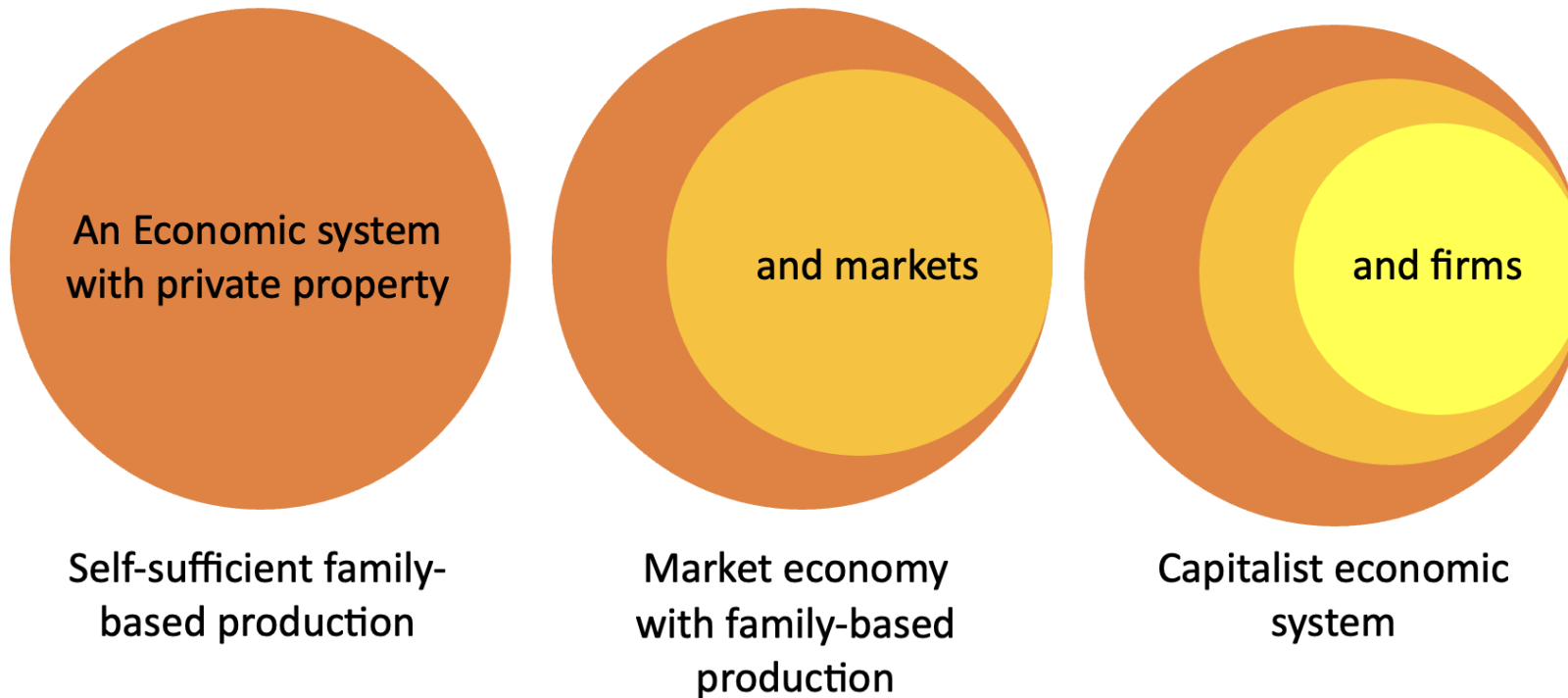
Defining capitalism

- Capitalism is an *economic system*.
- Capitalism is characterized by a new combination of *institutions* built around:
 - *Private property*
 - *Markets*
 - *Firms*
- But even capitalist economies rely on non-capitalist institutions
 - *Families*
 - *Government*



[1911 Industrial Workers of the World poster.](#)

Defining capitalism



In a capitalist system:

- Most production takes place in firms.
- Inputs and outputs are private property.
- Economic power is:
 - concentrated in the hands of owners and managers
 - limited by competition with other firms.

Figure 1.14 Capitalism: Private property, markets and firms.

How capitalism improved living standards

- *Specialisation*: The growth of firms and the expansion of markets allowed for an unprecedented division of labour.
- *Technology*: Firms competing with one another had strong incentives to adopt and develop new and more productive technologies.
- These ideas are usually attributed to Adam Smith.

GREAT ECONOMISTS

Adam Smith

Adam Smith (1723–1790) is considered by many to be the founder of modern economics. Raised by a widowed mother in Scotland, he went on to study philosophy at the University of Glasgow and later at Oxford, where he wrote: ‘the greater part of the ... professors have ... given up altogether even the pretence of teaching.’



He travelled throughout Europe, visiting Toulouse, France where he claimed to have ‘very little to do’ and thus began ‘to write a book in order to pass away the time.’ This was to become the most famous book in economics.

In *An Inquiry into the Nature and Causes of the Wealth of Nations*, published in 1776, Smith asked: how can society coordinate the independent activities of large numbers of economic actors—producers, transporters, sellers, consumers—often unknown to each other and widely scattered across the world? His radical claim was that coordination among all of these actors might spontaneously arise, without any person or institution consciously attempting to create or maintain it. This challenged previous notions of political and economic organization, in which rulers imposed order on their subjects.

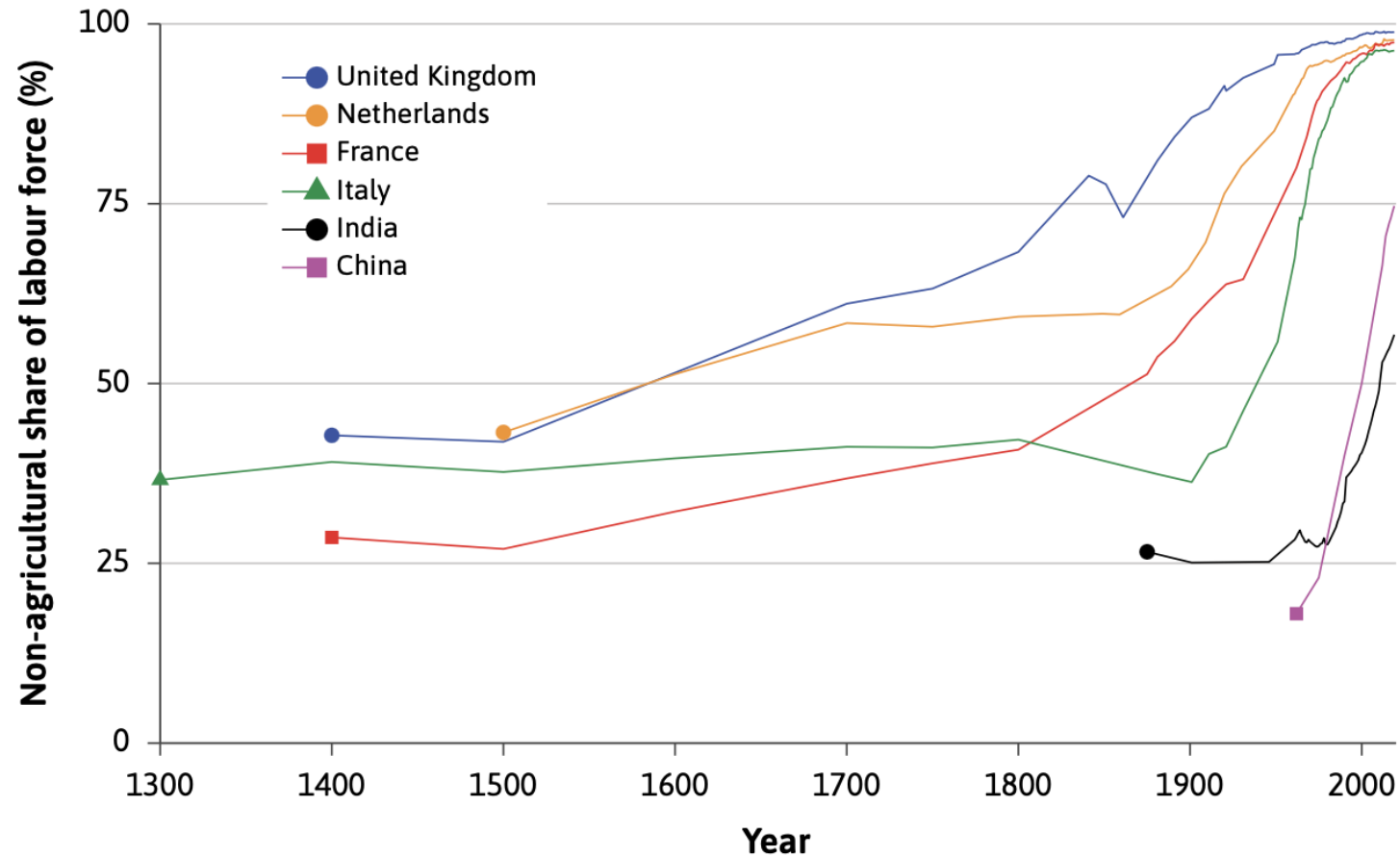
Gains from specialisation

- People are more productive when they focus on a limited range of activities due to:
 - Difference in ability
 - Learning by doing
 - Economies of scale
- People only specialise if they can acquire the other goods they need.
- Specialisation (or the division of labour) raises the question of how goods are distributed among people.



Farmers Boggis, Bunce, and Bean from *The Fantastic Mr. Fox* each represent a different aspect of specialization. Just as each farmer has their unique expertise—be it chickens, ducks, or apple cider—people in an economy are more productive when focusing on specific tasks.

Figure 1.15: Share of non-agricultural labour force in total labour force



Why less labour in agriculture?

- Arthur Lewis:
 - The economy has a subsistence sector on family farms and a modern capitalist sector producing industrial goods
 - There is a constant search for new technologies
 - This leads to increased production at a lower cost
 - Increase profits for owners of firms
 - Movement of workers out of the subsistence economy into capitalist sector (firms)
 - Before hockey stick tipped upwards there was an unlimited supply of labour in the subsistence sector.

Did capitalism cause the upward kink?

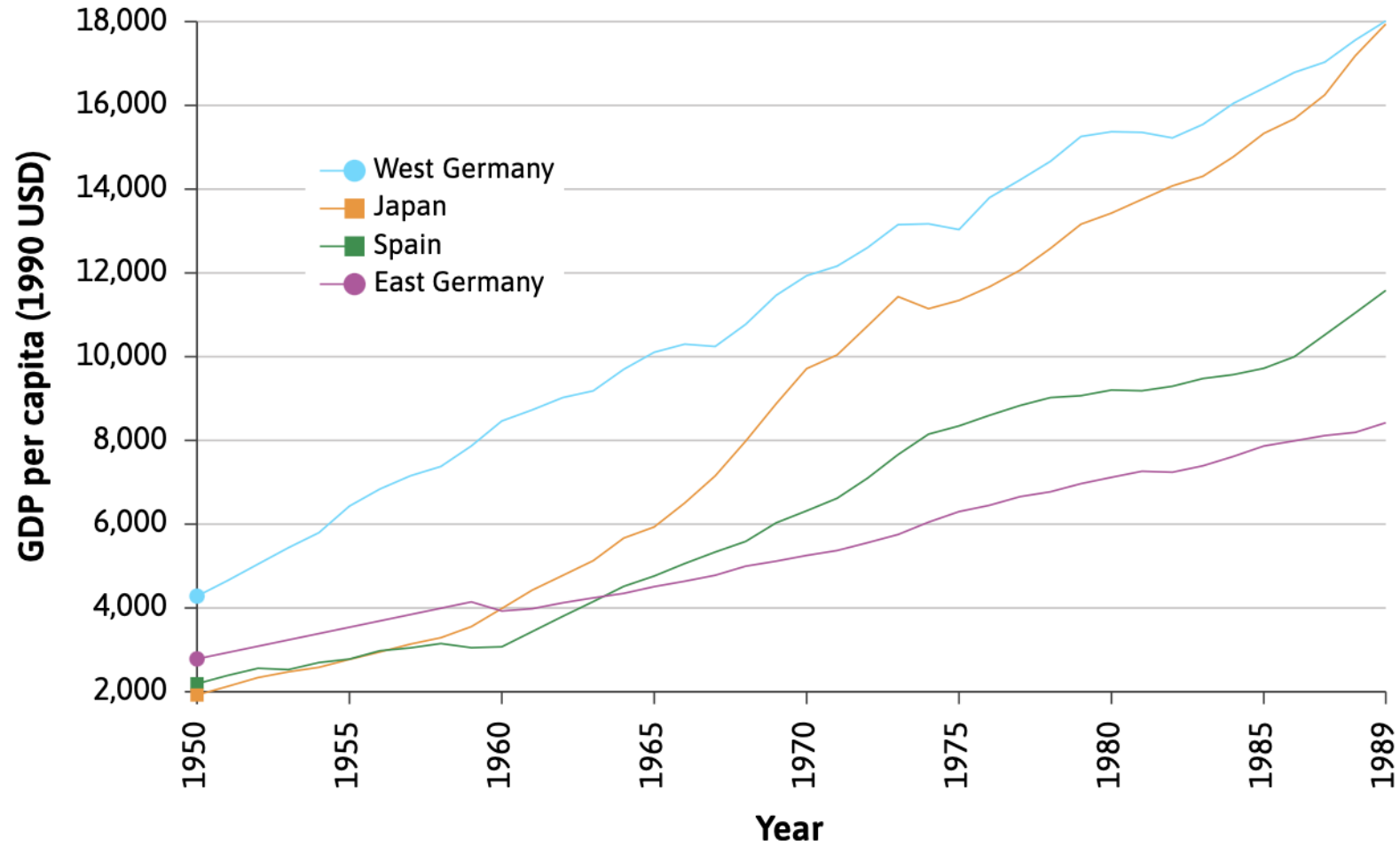
- A **causal statement** is one where we say that change X causes Y to happen.
- Scientists use experiments to test causal statements. This is difficult for economists. We make use of **natural experiments**.
- A **natural experiment** is a situation where we use an external event, such as a change in institutions or a natural disaster, to compare the outcomes for economic actors who were affected by this event and those who were not.
- A **counterfactual** event is what would have happened if there had been no change.

Capitalism, causation, and history's hockey sticks

- In science we support the statement that X *causes* Y by:
 - Understanding the relationship between cause and effect
 - Performing experiments to gather evidence by measuring X and Y
- Proving causation is difficult in economics
 - The economy is very complex!
 - We generally can't (or shouldn't) perform large scale experiments on entire populations.
- Capitalism emerged in Europe about the same time as the Enlightenment and the expansion of free speech and democracy.



Did capitalism cause the upward kink?



Colonization and economic performance

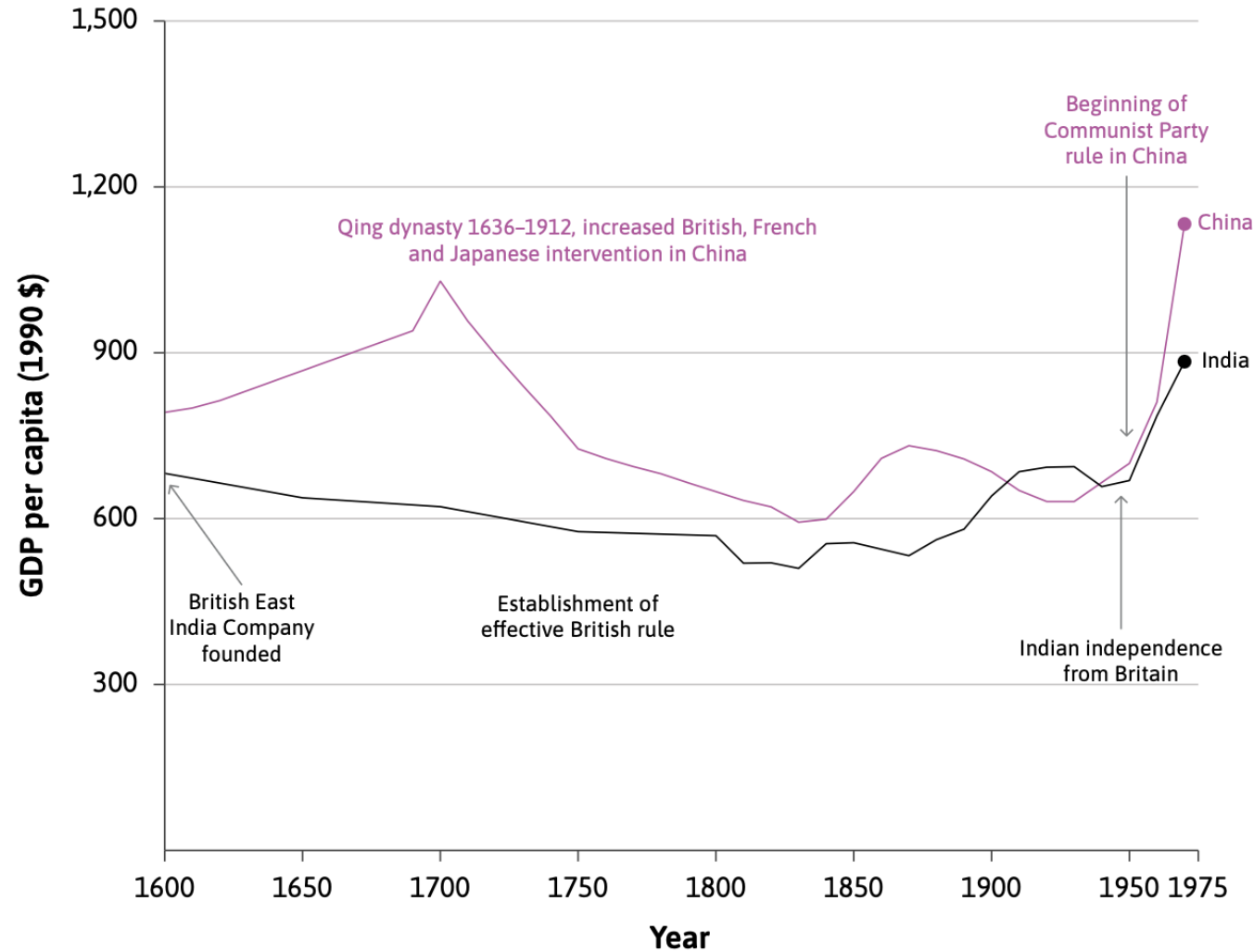


Figure 1.18: Shares of world output: 1500 - 2020

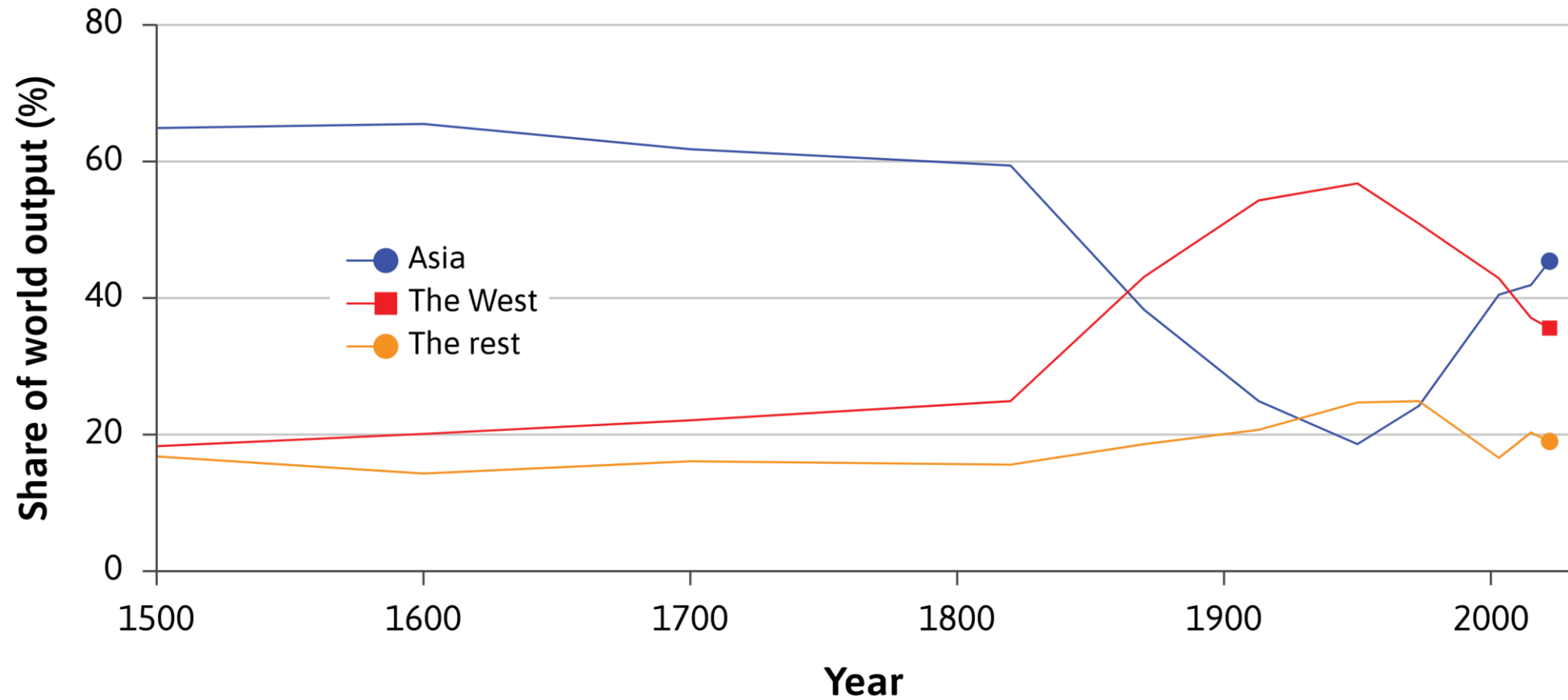
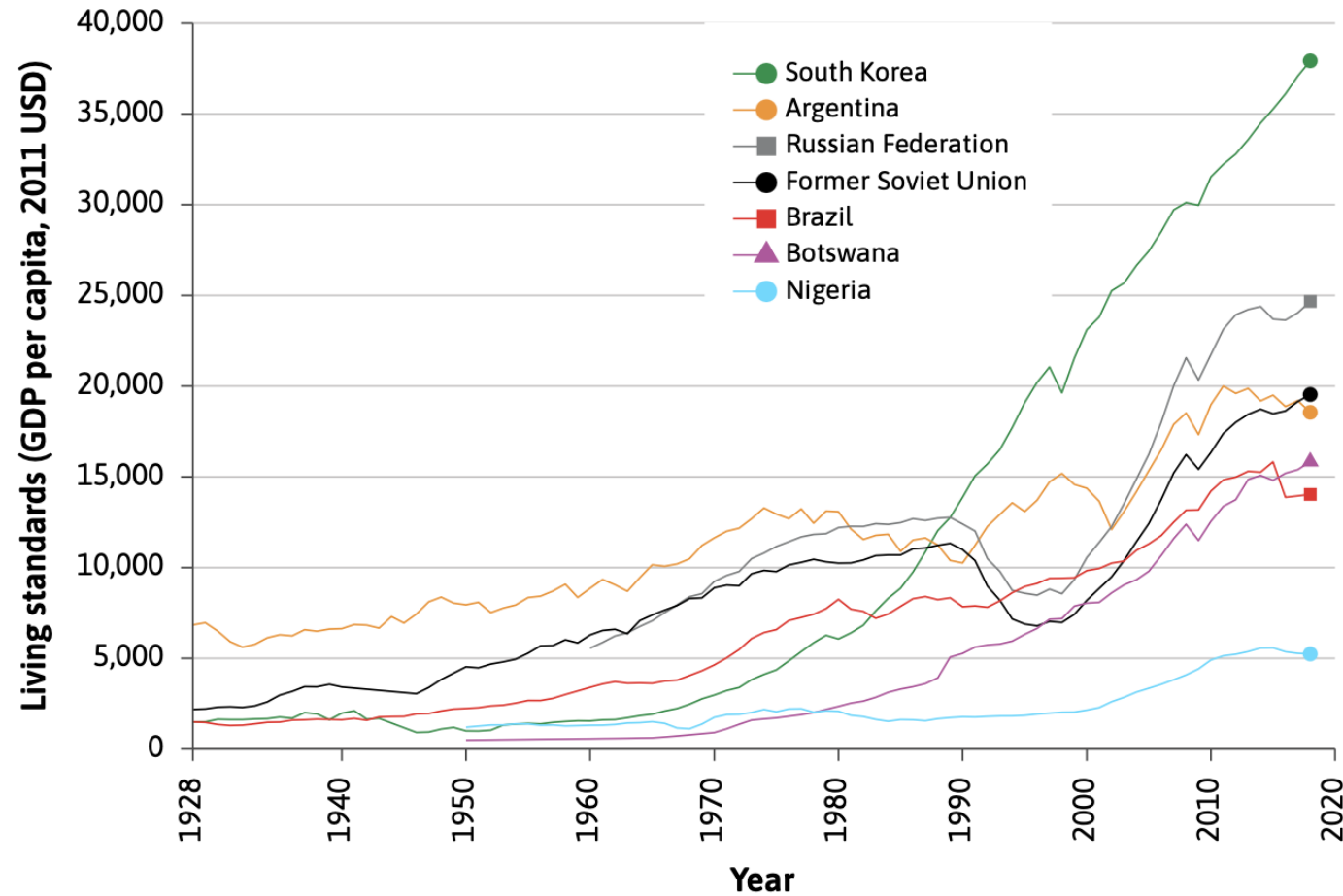
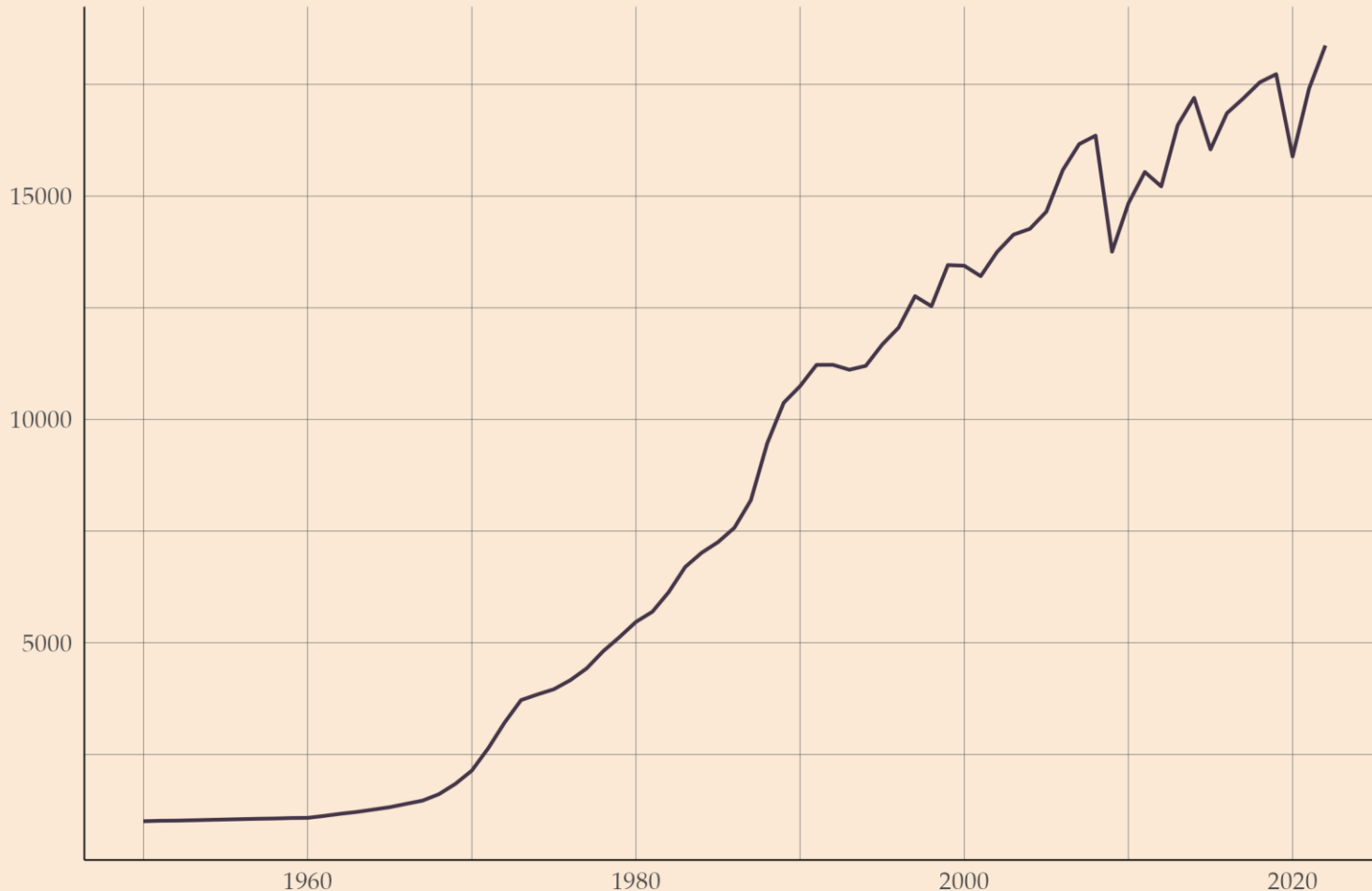


Figure 1.19: GDP per capita of latecomers to the capitalist revolution (1928-2018)



Botswana Hockey stick

Botswana GDP Per Capita
Constant US Dollar (\$) 2022



Data: World Inequality Database

- Botswana has maintained one of the highest per capita economic growth rates in the world over the past 50 years.
- 1966 at Independence 12 km of paved roads, 22 Botswana graduated from University and 100 from High School.
- Why Botswana?
- (Under Seretse Khama) Good “capitalist” policies formed the basis of the Botswana economy.
 - Institutions of private property.
- Limited colonial rule experienced by Botswana.

What is needed for capitalist dynamic?

- Economic conditions

- Private property must be secured
- Markets for firms must be competitive
- Firms are not controlled by people due to their connections to the government or privileged birth
- Markets, private property and firms are all regulated by laws and policies
- There is a well-functioning legal system, and the government provides essential goods and services

- Political conditions

- Capitalism and democracy frequently go hand in hand, but it is not a prerequisite for economic growth

The economy and the biosphere

- **Economics** is the study of how people interact with each other and with their natural environment in producing and acquiring their livelihoods, and how this changes over time and differs across societies.
- The **biosphere** is the collection of all forms of life on earth. The physical environment and the biosphere provide the essentials for life such as air, water and food as well as the raw materials that we use in production such as wood, metals and oil.

Figures 1.20 and 1.21: The economy

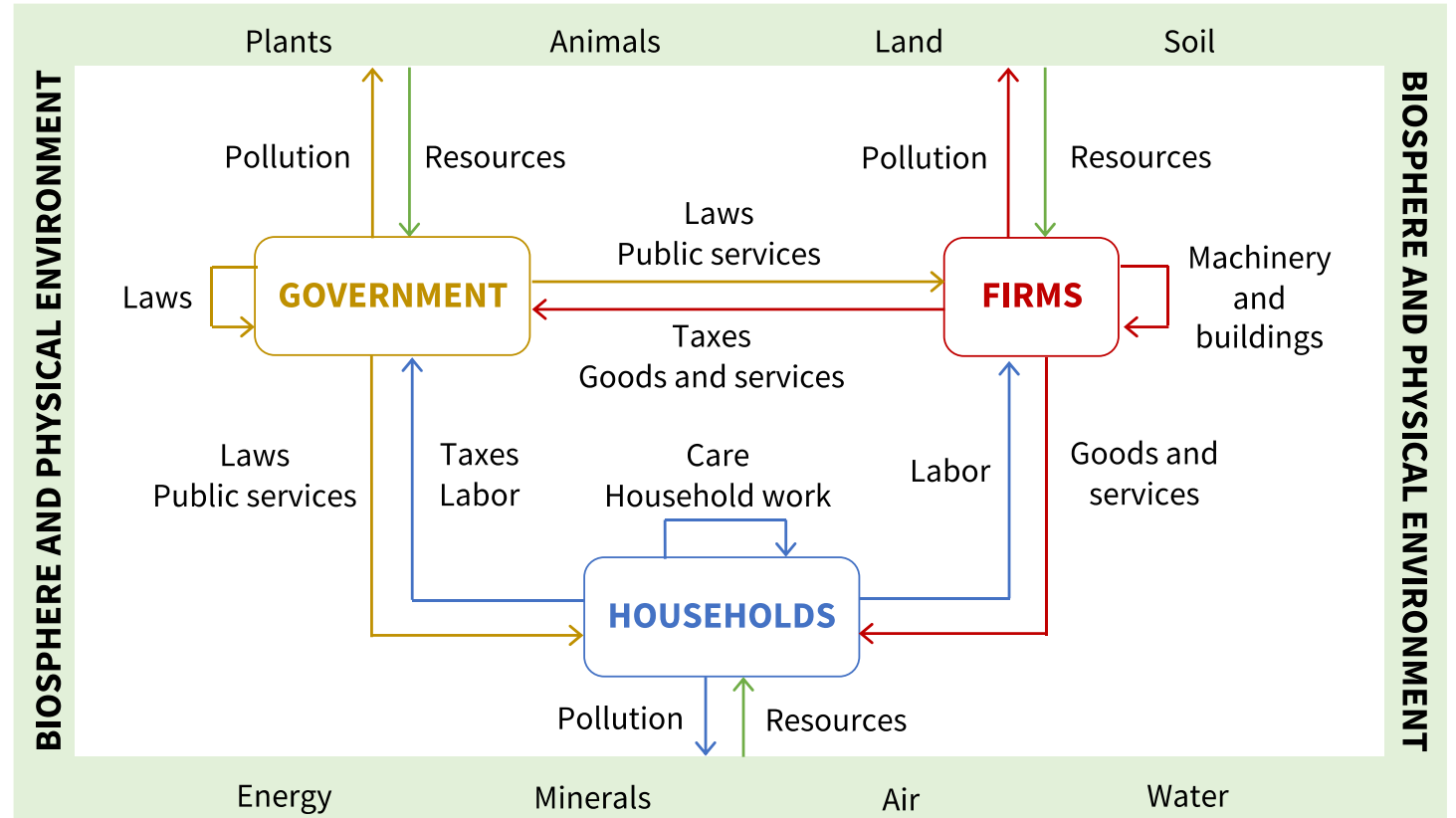
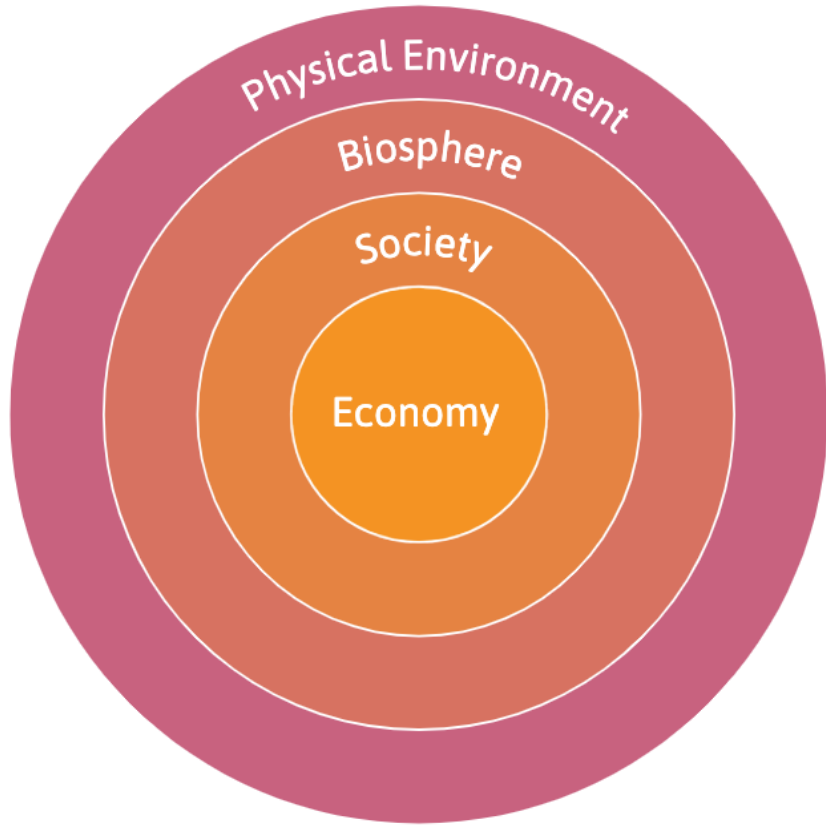
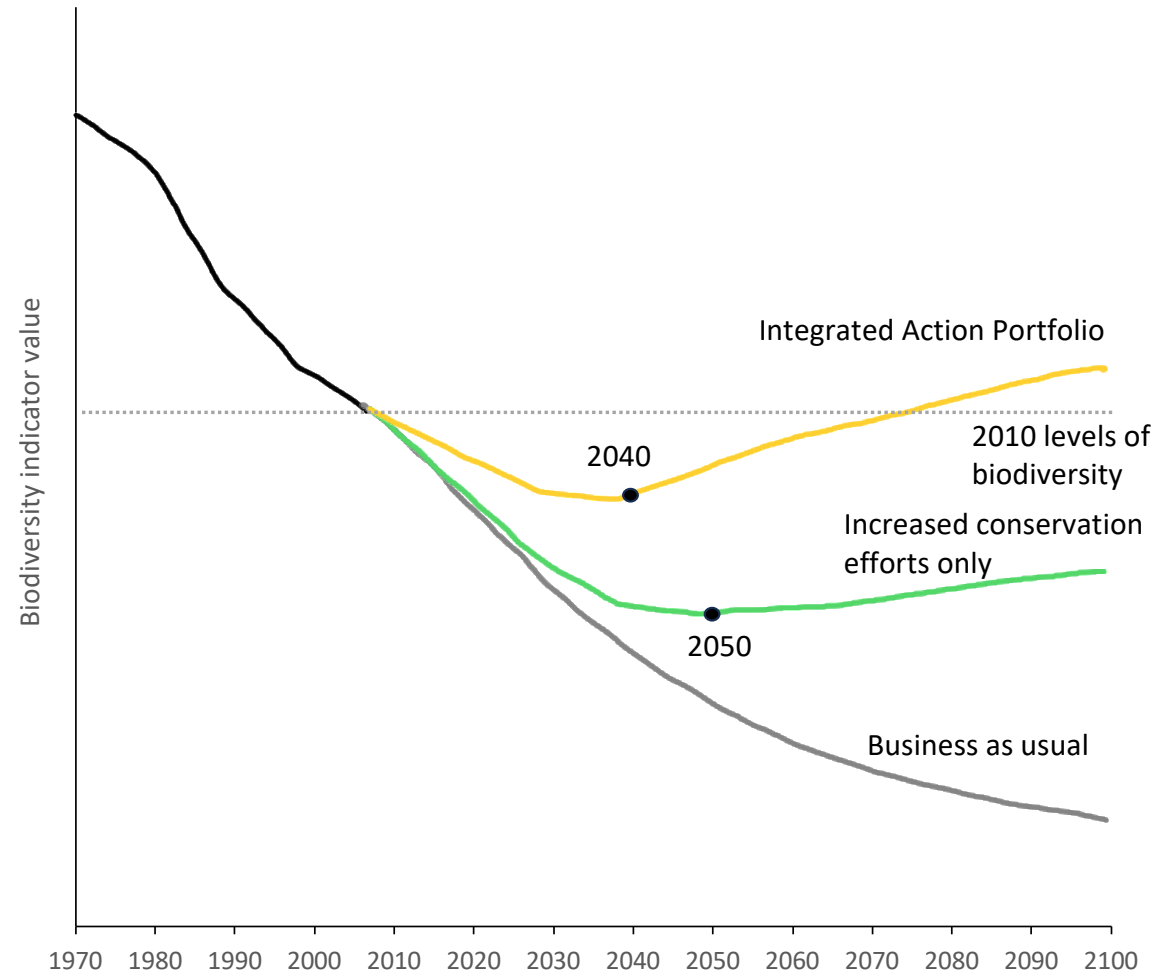


Figure 1.22: Global biodiversity loss under three scenarios



Summary

1. Important trends in economic variables over time
 - Income inequality across regions has increased a lot over time
 - “Hockey-stick” growth in GDP, and its negative consequences
 - **Technological progress** helped bring about these trends

2. The adoption of capitalism was another key factor
 - **Capitalism = Private property + Markets + Firms**
 - Failure of these institutions can explain divergence in economic growth across countries
 - Political systems and the role of government also determine the type of capitalist society

UNIT 1: Prosperity, Inequality, and Planetary Limits

1.2 History's Hockey Stick

1. Introduction: Rising Living Standards

- Living standards have risen dramatically since the time of **Ibn Battuta**.
 - However, the increase has:
 - Occurred much more in some countries than in others.
 - Happened at different times across regions.
-

2. Measuring Living Standards

2.1 Gross Domestic Product (GDP)

Definition of GDP

- **Gross Domestic Product (GDP):**
 - A measure of the **market value of final goods and services** produced in a country in a given period.
 - Includes output by:
 - Firms
 - Non-profit institutions
 - Government bodies
 - Excludes intermediate goods (to avoid double counting).
 - Includes household production if sold.
 - Measured monthly, quarterly, and annually.
 - GDP = Total output of a country in one year.
 - Economist **Diane Coyle** describes GDP as:

“Adding up everything from nails to toothbrushes, tractors, shoes, haircuts, management consultancy, street cleaning, yoga teaching, plates, bandages, books, and millions of other services and products.”
-

2.2 GDP Per Capita

- To measure **average living standards**, we:
 - Calculate total GDP.
 - Divide by total population.
- **GDP per capita = $\text{GDP} \div \text{Population}$**
 - Used as a proxy for average income.
 - Indicates average living standards.

(Note: Some important limitations are discussed later in the section.)

2.3 Purchasing Power Parity (PPP)

- Used to compare living standards across countries.
- Adjusts for differences in price levels between countries.

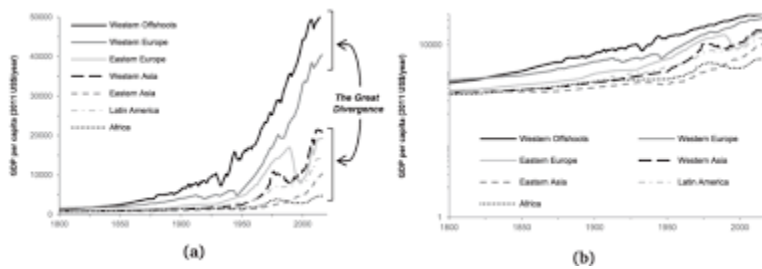
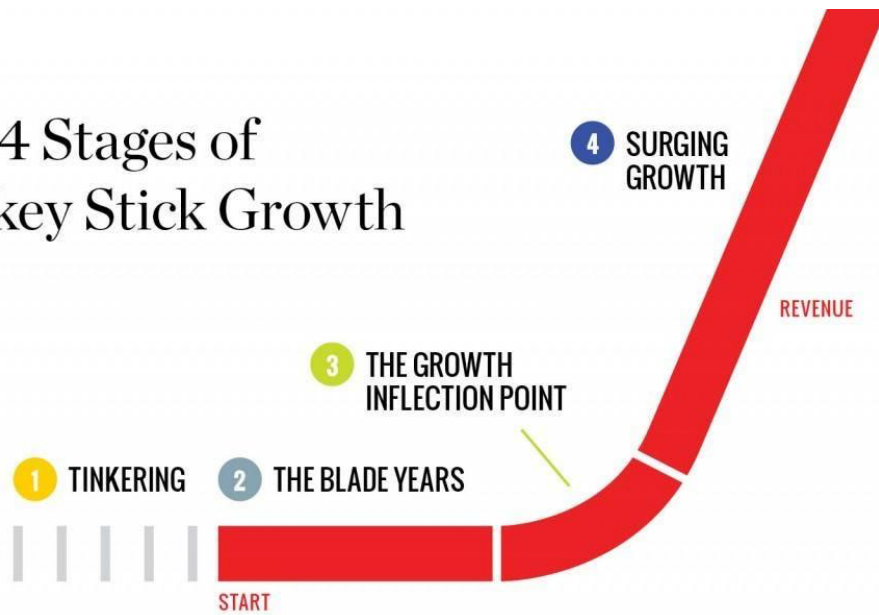
Definition:

- PPP measures:
 - The cost of purchasing a basket of goods and services in one country
 - Compared to the cost of the same basket in a reference country (e.g., US in 2011).
 - Expressed as price indices.
-

3. History's Hockey Stick (Figure 1.1)

 **Figure 1.1: GDP per capita in five countries (1000–2018)**

The 4 Stages of Hockey Stick Growth



4

Description of the Graph

- X-axis: Year (1000–2018)
- Y-axis: GDP per capita (2011 US\$)
- Countries shown:
 - Britain
 - Japan
 - Italy
 - China
 - India

Why “Hockey Stick”?

- The graph resembles an ice hockey stick:
 - Long period of flat growth.
 - Sharp upward curve toward the end.

4. Key Observations from Figure 1.1

4.1 Pre-Modern Period (Before ~1500)

- Very limited sustained growth.
 - In the 14th century:
 - Italy had the highest living standards among the five countries.
 - Data before 1300 is scarce:
 - For China, estimates exist only for 1000, 1090, and 1120.
 - Straight lines connect limited data points.
-

4.2 The “Kink” in Growth

- Sustained economic growth began at different times:

Country	Approximate Take-Off
Britain	~1650 (gradual kink)
Japan	~1870 (sharper kink)
China	Late 20th century
India	Late 20th century

- In China and India:
 - Living standards declined while Western Europe industrialised.
 - Growth accelerated much later.
-

4.3 Colonial Influence

- In some economies (China, India):
 - Substantial improvements occurred after:
 - Independence from colonial rule.
 - Reduced European interference.
-

4.4 Modern Differences (By 2018)

- Japan's GDP per capita $\approx$ 6 times India's.
 - Japan nearly as rich as Britain.
 - US (not shown) richer than Japan.
 - Norway (not shown) richer still.
-

4.5 Catch-Up Growth

- Since late 20th century:
 - "Latecomers" like China and India have been catching up.
 - In some countries:
 - The hockey stick has not yet tipped upward.
-

5. Data and Economic History

- The graph is possible due to the work of:
 - **Angus Maddison**
 - Dedicated career to historical GDP reconstruction.
 - Recent economic historians have refined these estimates.
 - Core message:
 - Economics begins with data about people and regions.
-

6. The Great Divergence

- Refers to:
 - The widening gap in living standards between Western Europe and the rest of the world.
 - Central question:
 - Why did some countries become rich earlier than others?
-

7. Adam Smith and the Wealth of Nations

7.1 Introduction

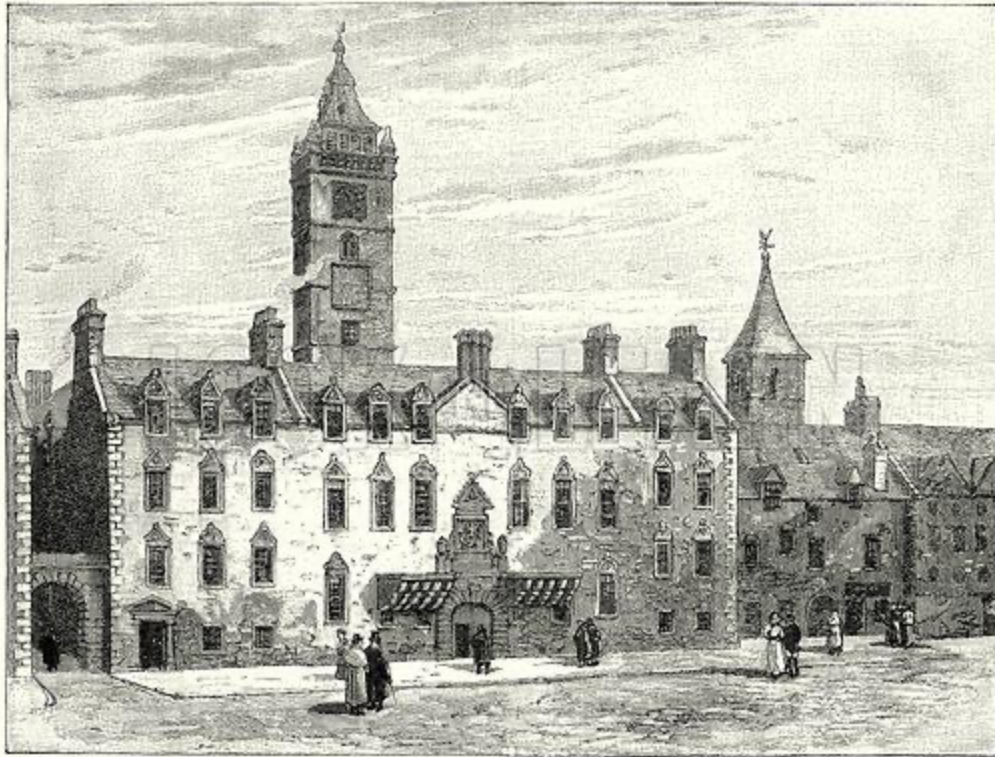


Adam Smith.

WEALTH OF
NATIONS

ADAM
SMITH

GREAT MINDS SERIES



GLASGOW UNIVERSITY IN THE EIGHTEENTH CENTURY.

4

- **Adam Smith**
 - Founder of modern economics.
 - Studied at University of Glasgow.
 - Influenced by Enlightenment ideas and colonial trade.

7.2 Major Work

- **An Inquiry into the Nature and Causes of the Wealth of Nations (1776)**

Central Question:

How can society coordinate the independent actions of many economic actors without central control?

7.3 Key Ideas

1. Spontaneous Order

- Coordination can arise:
 - Without deliberate planning.
 - Without a central authority.
-

2. Self-Interest

- Individuals pursuing self-interest can:
 - Promote society's welfare.
 - Famous quote:
"It is not from the benevolence of the butcher, brewer, or baker that we expect our dinner..."
-

3. The Invisible Hand

- Metaphor for unintended coordination.
 - Individuals seeking personal gain may:
 - Promote societal good.
-

4. Division of Labour

- Specialization increases productivity.
 - Example:
 - Pin factory:
 - 10 specialized workers → ~50,000 pins/day.
 - Individually → maybe 1 pin/day.
 - Constrained by:
 - "Extent of the market".
 - Larger markets allow greater specialization.
-

5. Role of Government

Smith supported:

- National defence.
- Justice system (courts + police).
- Public works (roads, canals, bridges).
- Education.

- Preventing monopolies.
-

7.4 Market Failures

- Smith warned:
 - Sellers may collude to raise prices.
 - Monopolies distort competition.
 - Criticized:
 - Government-protected monopolies such as the **British East India Company**.
-

7.5 Moral Sentiments

- Earlier book:
 - **The Theory of Moral Sentiments**
 - Shows:
 - Humans are not guided entirely by self-interest.
 - Ethical behaviour matters.
-

8. Core Lessons from History's Hockey Stick

- For most of history:
 - No sustained growth.
 - When growth began:
 - It started at different times across countries.
 - Result:
 - Massive global inequality.
 - Key economic question:
 - Why did sustained growth begin?
 - Why in some places first?
-

9. QUESTION 1.1 – Correct Answers

Correct options:

- ☒ 1. Adam Smith believed in the role of government to improve societal welfare.

- Defence, justice, public works, education.

✗ 2. Adam Smith believed all markets were perfectly competitive.

- Incorrect. He warned about monopolies and collusion.

✗ 3. Adam Smith argued agents were guided entirely by self-interest.

- Incorrect. He also emphasized moral sentiments.

✓ 4. Smith claimed coordination could arise spontaneously without central authority.

- This is the core of the invisible hand concept.

In the next unit

- Using economic models to explain the trends in technological growth over time
- The role of firms in technological development
- **Malthusian economics:** studying the interaction between population, technology, and economic growth

UNIT 1: Prosperity, Inequality, and Planetary Limits

1.2 History's Hockey Stick

1. Introduction: Rising Living Standards

- Living standards have risen dramatically since the time of **Ibn Battuta**.
 - However, the increase has:
 - Occurred much more in some countries than in others.
 - Happened at different times across regions.
-

2. Measuring Living Standards

2.1 Gross Domestic Product (GDP)

Definition of GDP

- **Gross Domestic Product (GDP):**
 - A measure of the **market value of final goods and services** produced in a country in a given period.
 - Includes output by:
 - Firms
 - Non-profit institutions
 - Government bodies
 - Excludes intermediate goods (to avoid double counting).
 - Includes household production if sold.
 - Measured monthly, quarterly, and annually.
 - GDP = Total output of a country in one year.
 - Economist **Diane Coyle** describes GDP as:

“Adding up everything from nails to toothbrushes, tractors, shoes, haircuts, management consultancy, street cleaning, yoga teaching, plates, bandages, books, and millions of other services and products.”
-

2.2 GDP Per Capita

- To measure **average living standards**, we:
 - Calculate total GDP.
 - Divide by total population.
- **GDP per capita = $\text{GDP} \div \text{Population}$**
 - Used as a proxy for average income.
 - Indicates average living standards.

(Note: Some important limitations are discussed later in the section.)

2.3 Purchasing Power Parity (PPP)

- Used to compare living standards across countries.
- Adjusts for differences in price levels between countries.

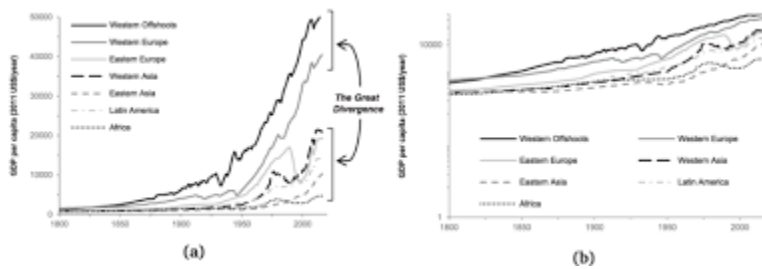
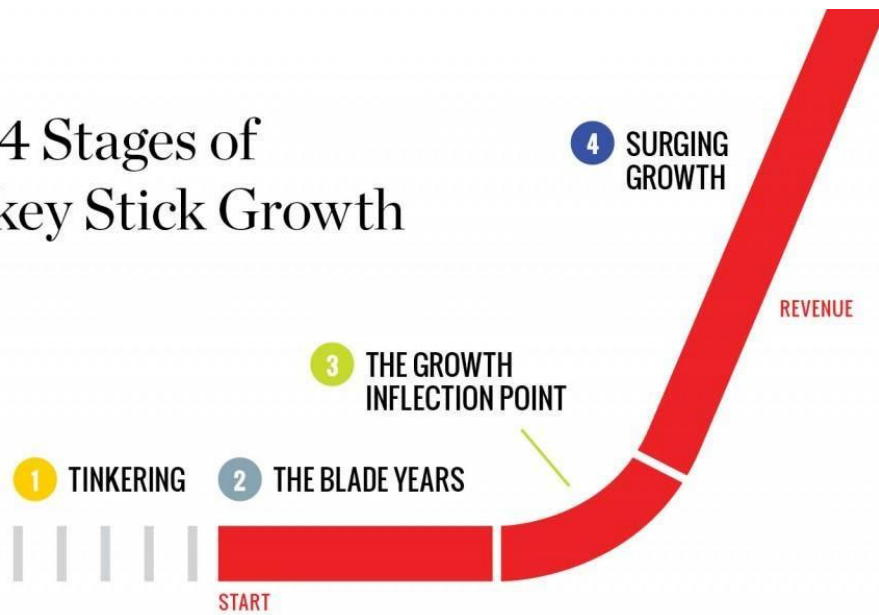
Definition:

- PPP measures:
 - The cost of purchasing a basket of goods and services in one country
 - Compared to the cost of the same basket in a reference country (e.g., US in 2011).
 - Expressed as price indices.
-

3. History's Hockey Stick (Figure 1.1)

 **Figure 1.1: GDP per capita in five countries (1000–2018)**

The 4 Stages of Hockey Stick Growth



4

Description of the Graph

- X-axis: Year (1000–2018)
- Y-axis: GDP per capita (2011 US\$)
- Countries shown:
 - Britain
 - Japan
 - Italy
 - China
 - India

Why “Hockey Stick”?

- The graph resembles an ice hockey stick:
 - Long period of flat growth.
 - Sharp upward curve toward the end.

4. Key Observations from Figure 1.1

4.1 Pre-Modern Period (Before ~1500)

- Very limited sustained growth.
 - In the 14th century:
 - Italy had the highest living standards among the five countries.
 - Data before 1300 is scarce:
 - For China, estimates exist only for 1000, 1090, and 1120.
 - Straight lines connect limited data points.
-

4.2 The “Kink” in Growth

- Sustained economic growth began at different times:

Country	Approximate Take-Off
Britain	~1650 (gradual kink)
Japan	~1870 (sharper kink)
China	Late 20th century
India	Late 20th century

- - In China and India:
 - Living standards declined while Western Europe industrialised.
 - Growth accelerated much later.
-

4.3 Colonial Influence

- In some economies (China, India):
 - Substantial improvements occurred after:
 - Independence from colonial rule.
 - Reduced European interference.
-

4.4 Modern Differences (By 2018)

- Japan's GDP per capita $\approx$ 6 times India's.
 - Japan nearly as rich as Britain.
 - US (not shown) richer than Japan.
 - Norway (not shown) richer still.
-

4.5 Catch-Up Growth

- Since late 20th century:
 - "Latecomers" like China and India have been catching up.
 - In some countries:
 - The hockey stick has not yet tipped upward.
-

5. Data and Economic History

- The graph is possible due to the work of:
 - **Angus Maddison**
 - Dedicated career to historical GDP reconstruction.
 - Recent economic historians have refined these estimates.
 - Core message:
 - Economics begins with data about people and regions.
-

6. The Great Divergence

- Refers to:
 - The widening gap in living standards between Western Europe and the rest of the world.
 - Central question:
 - Why did some countries become rich earlier than others?
-

7. Adam Smith and the Wealth of Nations

7.1 Introduction

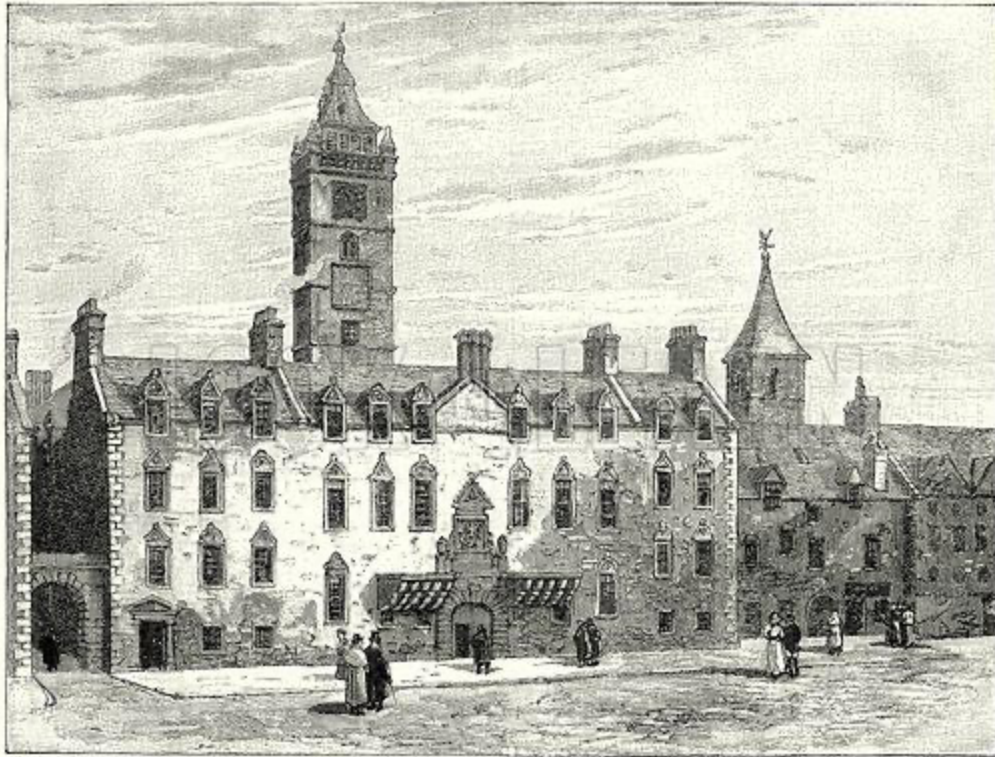


Adam Smith.

WEALTH OF
NATIONS

ADAM
SMITH

GREAT MINDS SERIES



GLASGOW UNIVERSITY IN THE EIGHTEENTH CENTURY.

4

- **Adam Smith**
 - Founder of modern economics.
 - Studied at University of Glasgow.
 - Influenced by Enlightenment ideas and colonial trade.

7.2 Major Work

- **An Inquiry into the Nature and Causes of the Wealth of Nations (1776)**

Central Question:

How can society coordinate the independent actions of many economic actors without central control?

7.3 Key Ideas

1. Spontaneous Order

- Coordination can arise:
 - Without deliberate planning.
 - Without a central authority.
-

2. Self-Interest

- Individuals pursuing self-interest can:
 - Promote society's welfare.
 - Famous quote:
"It is not from the benevolence of the butcher, brewer, or baker that we expect our dinner..."
-

3. The Invisible Hand

- Metaphor for unintended coordination.
 - Individuals seeking personal gain may:
 - Promote societal good.
-

4. Division of Labour

- Specialization increases productivity.
 - Example:
 - Pin factory:
 - 10 specialized workers → ~50,000 pins/day.
 - Individually → maybe 1 pin/day.
 - Constrained by:
 - "Extent of the market".
 - Larger markets allow greater specialization.
-

5. Role of Government

Smith supported:

- National defence.
- Justice system (courts + police).
- Public works (roads, canals, bridges).
- Education.

- Preventing monopolies.
-

7.4 Market Failures

- Smith warned:
 - Sellers may collude to raise prices.
 - Monopolies distort competition.
 - Criticized:
 - Government-protected monopolies such as the **British East India Company**.
-

7.5 Moral Sentiments

- Earlier book:
 - **The Theory of Moral Sentiments**
 - Shows:
 - Humans are not guided entirely by self-interest.
 - Ethical behaviour matters.
-

8. Core Lessons from History's Hockey Stick

- For most of history:
 - No sustained growth.
 - When growth began:
 - It started at different times across countries.
 - Result:
 - Massive global inequality.
 - Key economic question:
 - Why did sustained growth begin?
 - Why in some places first?
-

9. QUESTION 1.1 – Correct Answers

Correct options:

- ☒ 1. Adam Smith believed in the role of government to improve societal welfare.

- Defence, justice, public works, education.

✗ 2. Adam Smith believed all markets were perfectly competitive.

- Incorrect. He warned about monopolies and collusion.

✗ 3. Adam Smith argued agents were guided entirely by self-interest.

- Incorrect. He also emphasized moral sentiments.

✓ 4. Smith claimed coordination could arise spontaneously without central authority.

- This is the core of the invisible hand concept.